

Instructions to create automata (written both for a human & AI)

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Our aim is to construct an automaton, by which we will mean a directed labeled graph. Because the labels on the vertices of the automaton are themselves graphs, to reduce confusion, we will always call the vertices of the automaton “nodes” and reserve this terminology for the vertices of the automaton.

For us, a graph will always be allowed to have single-edge loops and multi-edges. Sometimes it will be directed, and sometimes it will have labels on the vertices and edges.

Admissible ltt structures

The first goal is to write code to list all admissible ltt structures (these will give potential nodes of the automaton). The following definitions build up to defining an admissible ltt structure.

We are interested in a specific kind of colored graph that will have single-edge loops and multi-edges. We provide here this definition and henceforth refer to it as a structure graph:

Definition 0.1 (structure graph). By a *structure graph* we mean a graph in which each of (1)–(4) holds:

- 1) There is exactly one red vertex; all other vertices are purple.
- 2) There is exactly one red edge; all other edges are either black or purple. Edges colored red or purple are called *colored edges*.
- 3) Multi-edges are allowed, but for each pair of vertices there is at most one colored edge connecting that pair.
- 4) Single-edge loops are allowed, but only black edges can be loops.

In practice, a structure graph will have an “underlying graph” with 5-edge that a colored graph, as in Definition 0.1 is obtained from as a “blow up.” The edges of the underlying graph become 5 black edges and we label these black edges by $x_1, \dots, x_5$. We further assigned an orientation to the 5 black edges, labeling the initial vertex of the black edge x_j by x_j and the terminal vertex of x_j by $\bar{x}_j$. A bar over an edge name then also indicates a reversal of orientation. With this in mind, the following notation is used to discuss the black edges and vertices:

Definition 0.2 (notation). For simplicity, we establish notation for this list of vertex labels:

Define

$$\mathcal{L} = \{x_1, \bar{x}_1, \dots, x_5, \bar{x}_5\}.$$

And we notationally encode that reversing the orientation of an edge twice returns the original edge:

$$\bar{\bar{x}}_j = x_j.$$

Finally, in order to make choices about the order in which we perform certain computations, we will need an order on the set of vertex labels $\mathcal{L}$:

Assume

$$x_1 < \bar{x}_1 < \dots < x_5 < \bar{x}_5$$

is the increasing order on $\mathcal{L}$.

Definition 0.3 (distinguished admissible ltt structure). A *distinguished admissible ltt structure* is a connected structure graph G with:

- 5 pairs of vertices labeled precisely by the elements of $\mathcal{L}$ (duplicate labels are not allowed);
- x_1 is red and all remaining vertices are purple;

- x_1 is connected by a red edge to x_2 ;
- $\{\bar{x}_1, x_2, \bar{x}_2, \dots, x_5, \bar{x}_5\}$ is partitioned into three sets of size 3, and vertices in each partition element are pairwise connected by purple edges;
- each pair $\{x_j, \bar{x}_j\}$ is connected by a black edge (this black edge can be viewed as an oriented edge from $\{x_j$ to $\bar{x}_j\}$, and then $\bar{x}_j\}$ would notate this edge with a reverse orientation).

Definition 0.4 (ep-isomorphic, admissible ltt structure). Two colored graphs G_1, G_2 with vertices labeled by $\mathcal{L}$ are *ep-isomorphic* via bijection φ iff there is a bijection $\varphi : \mathcal{L} \rightarrow \mathcal{L}$ such that:

- (i) for each $x \in \mathcal{L}$, the vertices labeled x and $\varphi(x)$ have the same color;
- (ii) for each distinct pair of elements $x, y \in \mathcal{L}$, the vertices labeled $\varphi(x), \varphi(y)$ are connected by an edge of color C iff the vertices labeled x, y are connected by an edge of color C (black here is considered to be a color).

An *admissible ltt structure* is any graph ep-isomorphic to a distinguished admissible ltt structure. The red vertex $\varphi(x_1)$ will sometimes be denoted by v_r , the other red edge vertex $\varphi(x_2)$ will sometimes be denoted by v_p , and the red edge oriented (v_r, v_p) will sometimes be denoted by e_r . The set of purple edges may be denoted by $\mathcal{P}$.

Place some (any) order on the set of admissible ltt structures. Let PLTT denote the set of admissible ltt structures endowed with this order.

Note that the partition of $\{\bar{x}_1, x_2, \bar{x}_2, \dots, x_5, \bar{x}_5\}$ induces a partition of the set of purple edges. Recall that there is an ordering on $\mathcal{L}$ - we will use this to order the partition elements. In the partition of both $\mathcal{L}$ and $\mathcal{P}$, we order the partition elements in the increasing order of their minimal element. That is, we denote the partition elements by L_1, L_2, L_3 , where $i < j$ when there is an $\ell \in L_i$ such that $\ell < \ell'$ for each $\ell' \in L_j$. The indices of the partition $\mathcal{P} = \{P_1, P_2, P_3\}$ is consistent. That is, the partition elements are ordered by their lowest vertex labels.

Recall that an admissible ltt structure G has an underlying graph, we denote Γ_G . In practice, this underlying graph can be obtained by collapsing each colored edge to a point. As such, there is a bijection between the vertex set of Γ_G and the partition of $\mathcal{P}$ (and equivalently of $\mathcal{L}$). Denote by v_j the vertex associated to the partition element P_j .

Convention 0.5. An admissible ltt structure will be described in the following format:

$$e_r: (v_r, v_p) \quad v_1: P_1 \quad v_2: P_2 \quad v_3: P_3$$

Task 0.6. Write code to list all admissible ltt structures. Print out a representative of each ep-isomorphism class.

Output in the appendix.

Task 0.7. Create a visual depiction of the admissible ltt structures constructed in Task 0.6.

Output in the appendix.

1 Algorithm for constructing automata

Before constructing the final automaton, we will need to construct a directed labeled graph that we will call preAUT. The nodes of preAUT will be certain elements of PLTT (the set of all nodes will be notated *GRAPHS* and the directed edges connecting them will have labels that are also determined by the following algorithm. After constructing preAUT, we will take the disjoint union of the maximal strongly connected components, which will give us the automaton (directed labeled graph) AUT. AUT will be the final output of the program that we are constructing.

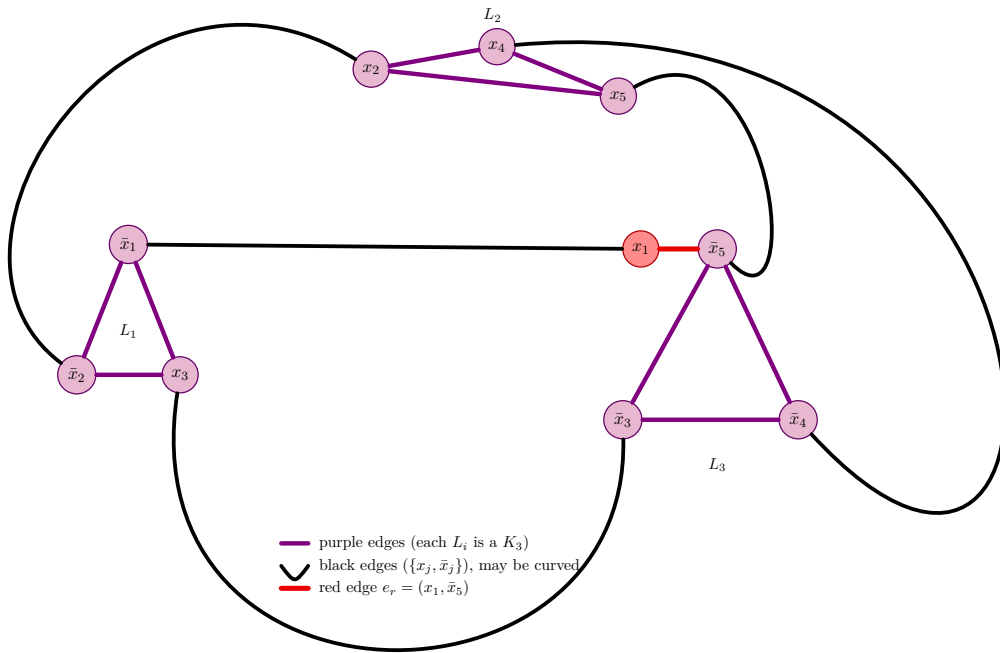
1.1 Base Step

For us the *seed ltt structure* will be that given by:

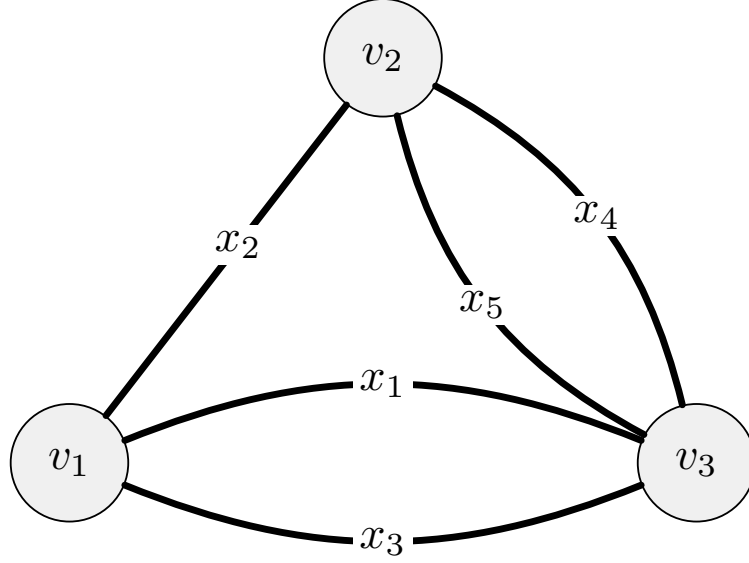
$$\begin{aligned} e_r: & (x_1, \bar{x}_5) \\ v_1: & \{\bar{x}_1, \bar{x}_2\}, \{\bar{x}_2, x_3\}, \{\bar{x}_1, x_3\} \\ v_2: & \{x_2, x_4\}, \{x_4, x_5\}, \{x_2, x_5\} \\ v_3: & \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_4, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\} \end{aligned}$$

Task 1.1. Create a visual depiction of the seed ltt structure.

Task 1.2. Create a visual depiction of the underlying graph of the seed ltt structure.



Vertices follow Convention 0.5: x_1 is the unique red vertex; $\bar{x}_5$ is the purple endpoint of e_r ; each triple L_i spans a purple clique; each $\{x_j, \bar{x}_j\}$ is a black edge.



Here v_i corresponds to collapsing the purple clique on the labels in L_i , and collapsing the red edge identifies the red vertex x_1 with $\bar{x}_5 \in L_3$, so the black edge $\{x_1, \bar{x}_1\}$ becomes a strand between v_3 and v_1 labeled x_1 ; the other black edges are read off from which L_a contains x_j and which contains $\bar{x}_j$.

Step 1. Start with a particular ltt structure G . Add a node to preAUT that is labeled by G . For the first component of the automaton, we start with the seed ltt structure.

- $\text{GRAPHS} = \{G\}$;
- $\text{graphs}_1 = \{G\}$;
- $\text{boxes} = \{\text{graphs}_1\}$;
- layer 1 consists only of G , i.e. $\text{GRAPHS}_1 = \{G\}$.

Remark. GRAPHS will ultimately be the list of all nodes in the automaton.

- each graphs_k will be a “box” of ep-isomorphic nodes;
- boxes will be the collection of these ep-isomorphism classes;
- before closing loops, where nodes represent the same ep-isomorphism class of ltt structures, the automaton looks like:
 - each GRAPHS_2 (layer 2) node is the terminal node of a directed edge from the GRAPHS_1 (layer 1) node
 - each GRAPHS_3 (layer 3) node is the terminal node of a directed edge from at least one GRAPHS_2 (layer 2) node
 - in general, each GRAPHS_{k+1} (layer $k+1$) node is the terminal node of a directed edge from at least one GRAPHS_k (layer k) node

1.2 Recursive Step

Recursively (with respect to k), complete in PLTT order Steps 2–3 for each $G \in \text{GRAPHS}_k$.

1.2.1 Step 2 (determining outgoing edges & terminal graphs)

We find the 4 outgoing edges (terminal ltt structures out of G) as follows:

- Input the starting ltt structure G (an admissible ltt structure): with the vertices labeled by $\mathcal{L}$ and following Convention 0.5, but writing d_r for the label of the red vertex (instead of v_r) and d_p be for the label of the purple endpoint of the red edge (instead of v_p).
- There are exactly two vertices connected to the d_p -vertex by purple edges:
 - 1) let d_1 be the label of the first;
 - 2) let d_2 be the label of the second;

with order inherited from $\mathcal{L}$.

- For each of d_1, d_2 , there are two potential directed edges out of G (hence one potential terminal ltt structure for each such edge). The triples

(initial ltt structure, edge, terminal ltt structure)

will be written as (G, E_{jA}, G_{jA}) for each $j \in \{1, 2\}$ and $A \in \{a, b\}$, where the triples are determined as follows:

For $j \in \{1, 2\}$:

(G, E_{ja}, G_{ja}) :

The outgoing edge E_{ja} is labeled by $d_r \rightarrow d_j d_r$.

The terminal ltt structure G_{ja} has:

- red vertex d_r and red edge $e_r = (d_r, \bar{d}_j)$;
- precisely these purple edges:
 - * a purple edge $\{d, d'\}$ whenever G has a purple edge $\{d, d'\}$.

(G, E_{jb}, G_{jb}) :

The outgoing edge E_{jb} is labeled by $d_j \rightarrow d_r d_j$.

The terminal ltt structure G_{jb} has:

- red vertex d_j and red edge $e_r = (d_j, \bar{d}_r)$;
- precisely these purple edges:
 - * a purple edge $\{d_r, d\}$ whenever G has a colored edge $\{d_j, d\}$ and
 - * a purple edge $\{d, d'\}$ whenever G has a purple edge $\{d, d'\}$ and neither $d = d_r$ nor $d' = d_r$.

Let

$$\text{terminal}(G) := \{G_{1;a}, G_{1;b}, G_{2;a}, G_{2;b}\}$$

(repetitions allowed).

- Remove any graph from $\text{terminal}(G)$ that is not in PLTT.
- Each graph in $\text{terminal}(G)$ is considered to be of level $k + 1$.
- $\text{terminal}(G)$ and level $k + 1$ both inherit their order from PLTT.

Task 1.3. Run Step 2 for the seed ltt structure. Print the outcome.

Output in the appendix.

1.2.2 Step 3 (Updating preAUT and the sets)

In PLTT order, apply the following to each $G' \in \text{terminal}(G)$.

- If G' is ep-isomorphic to some $G'' \in \text{GRAPHS}$, but $G' \neq G''$:
 - $\{G'\}$ is added to GRAPHS , and
 - $\{G'\}$ is added to graphs_j where $G'' \in \text{graphs}_j$, and
 - add to preAUT a “permutation edge” $G' \rightarrow G''$ labeled by φ , where φ is the permutation of $\mathcal{L}$ from the ep-isomorphism, and add to preAUT a directed edge $G \rightarrow G''$ with the edge label being that for $G \rightarrow G'$ given as in Step 2.
- If there exists $G'' \in \text{GRAPHS}$ with $G' = G''$: add to preAUT a directed edge $G \rightarrow G''$ with the edge label being that for $G \rightarrow G'$ given as in Step 2.
- For the remaining $G' \in \text{terminal}(G)$:
 - $\{G'\}$ is added to GRAPHS and as a node to preAUT, and
 - $\text{graphs}_{i+1} = \{G'\}$ where $\text{graphs}_i \in \text{boxes}$ and $\text{graphs}_{i+1} \notin \text{boxes}$, and
 - $\{G'\}$ is added to GRAPHS_{k+1} , and
 - add to preAUT a directed edge $G \rightarrow G'$ with the edge label being that for $G \rightarrow G'$ given as in Step 2.

Task 1.4. Having completed running Step 2 for the seed ltt structure, run Step 3 on the output. Print the outcome in a manner that is easily readable by a human mathematician.

Output in the appendix.

1.2.3 Step 4 (level shift & program termination)

After Steps 2–3 are completed for each $G \in \text{GRAPHS}_k$, check whether $\text{GRAPHS}_{k+1} = \emptyset$.

If $\text{GRAPHS}_{k+1} = \emptyset$, terminate.

If $\text{GRAPHS}_{k+1} \neq \emptyset$, perform a level shift by replacing k with $k + 1$ (equivalently, take the current level to be $\text{GRAPHS}_k := \text{GRAPHS}_{k+1}$), and then return to Steps 2–3 for each $G \in \text{GRAPHS}_k$.

In other words, the recursive loop is:

For a fixed k :

run Steps 2–3 on all $G \in \text{GRAPHS}_k \rightarrow$ run Step 4 $\rightarrow$ if nonempty, set $k \leftarrow k+1$ and repeat.

1.2.4 Step 5

Output the preAUT according to this seed graph as nodes and edges (including ltt structure/map info), leaving out any components with no edges.

1.2.5 Step 6

If every ep-isomorphism class of admissible ltt structures is represented by a node in some component of preAUT constructed thus far, then take the disjoint union of these components as preAUT and move onto Step 7. Otherwise, run Step 1– Step 6 using a seed graph from an ep-isomorphism class that is not yet represented by any node.

1.2.6 Step 7

Output the disjoint union of the maximal strongly connected components of preAUT as AUT, again leaving out any components with no edges.

A Figures for Task 0.6 (ep-isomorphism class representatives)

The Python code `list_admissible_ltt_classes.py` lists all 270 connected distinguished admissible ltt structures; they fall into 14 ep-isomorphism classes. Below, for each class, one representative is drawn in the same graphic style as the seed ltt structure earlier in this document (purple triangles for the three partition elements L_1, L_2, L_3 , black edges for pairs $\{x_j, \bar{x}_j\}$, red edge (x_1, x_2)). The caption line records the distinguished indices returned by that program for graphs in the class.

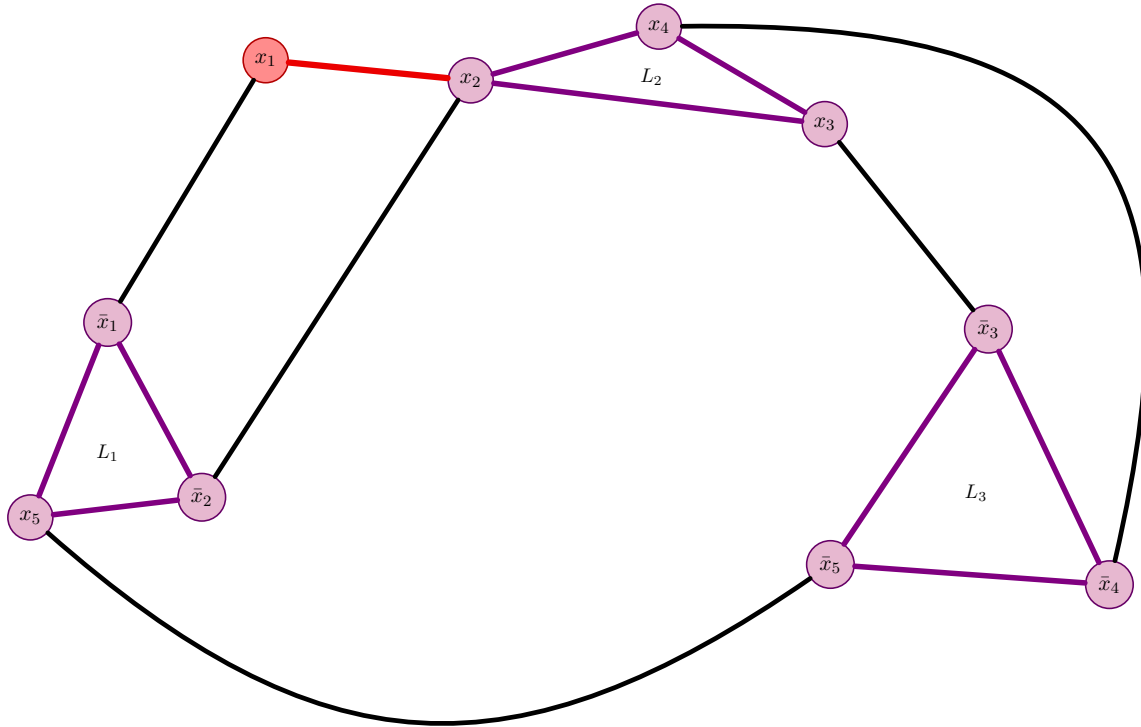
Regenerating the TikZ blocks. The file `ltt_appendix_generated.tex` is produced by running `python generate_ltt_appendix_tex.py` from the repository root.

Class 1 (II)

Representative of this ep-isomorphism class.

Distinguished indices:

$\{1, 8, 11, 19, 53, 60, 63, 65, 71, 80, 103, 110, 124, 125, 154, 160, 164, 165, 241, 244, 251, 257, 261, 268\}$,
24 total.

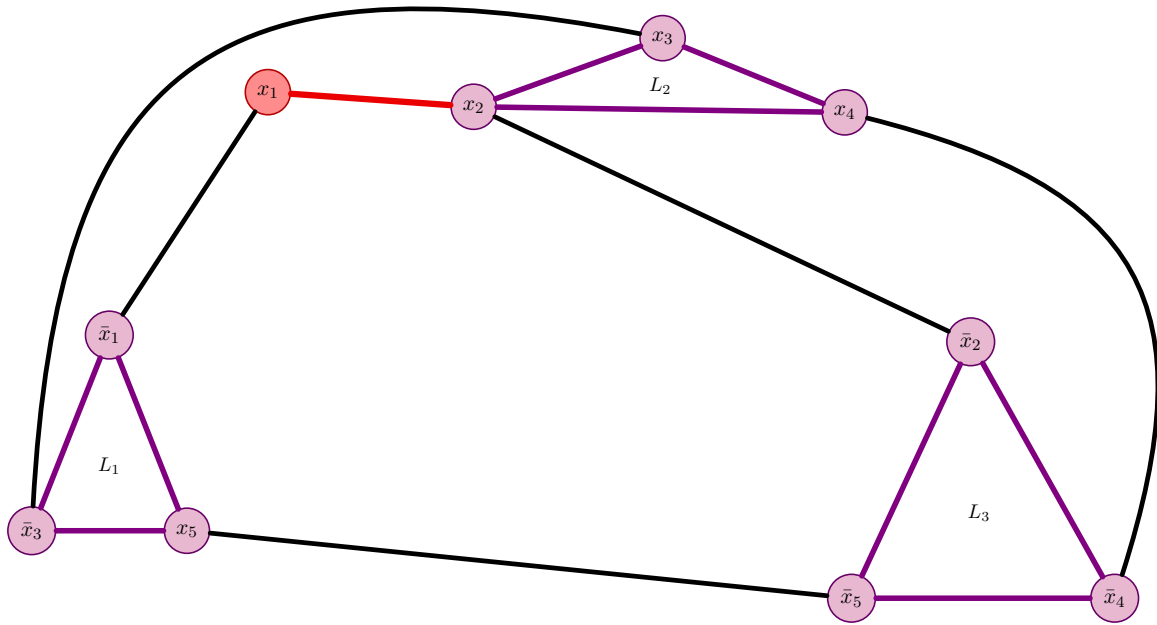


Class 2 (I)

Representative of this ep-isomorphism class.

Distinguished indices:

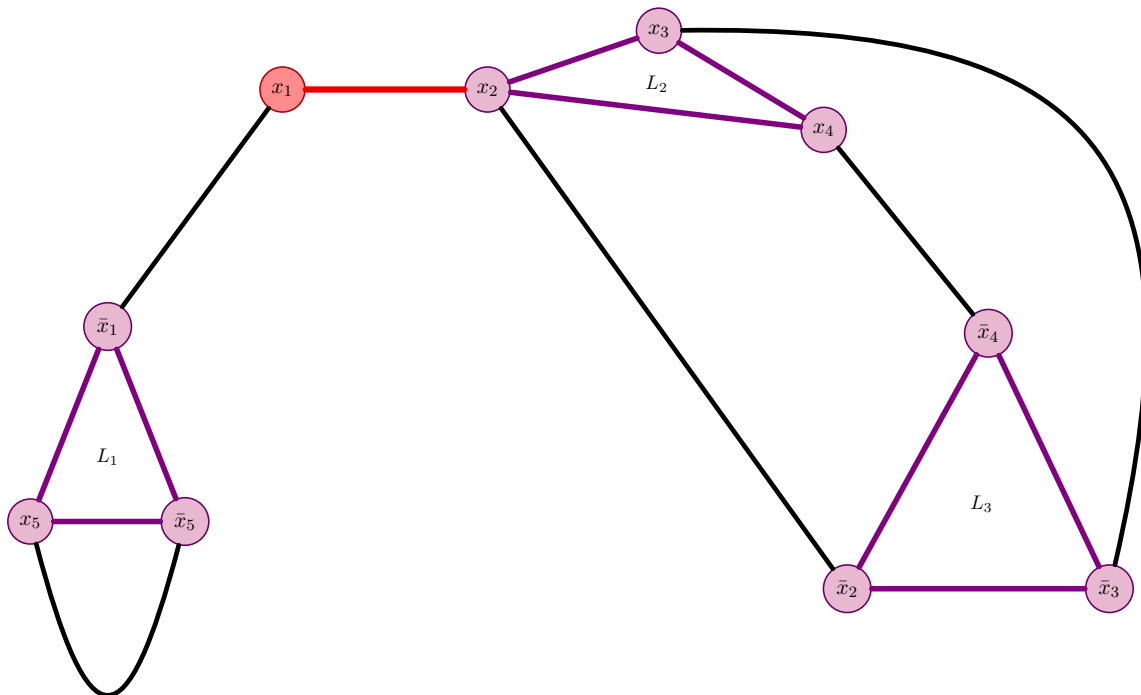
$\{2, 3, 5, 6, 12, 14, 15, 17, 51, 52, 57, 59, 61, 62, 66, 68, 73, 74, 76, 77, 101, 102, 107, 109, 121, 122, 127, 129, 151, 152, 156, 158, 161, 162, 167, 169, 245, 246, 249, 250, 252, 253, 259, 260, 262, 263, 265, 266\}$,
48 total.



Class 3 (VII)

Representative of this ep-isomorphism class.

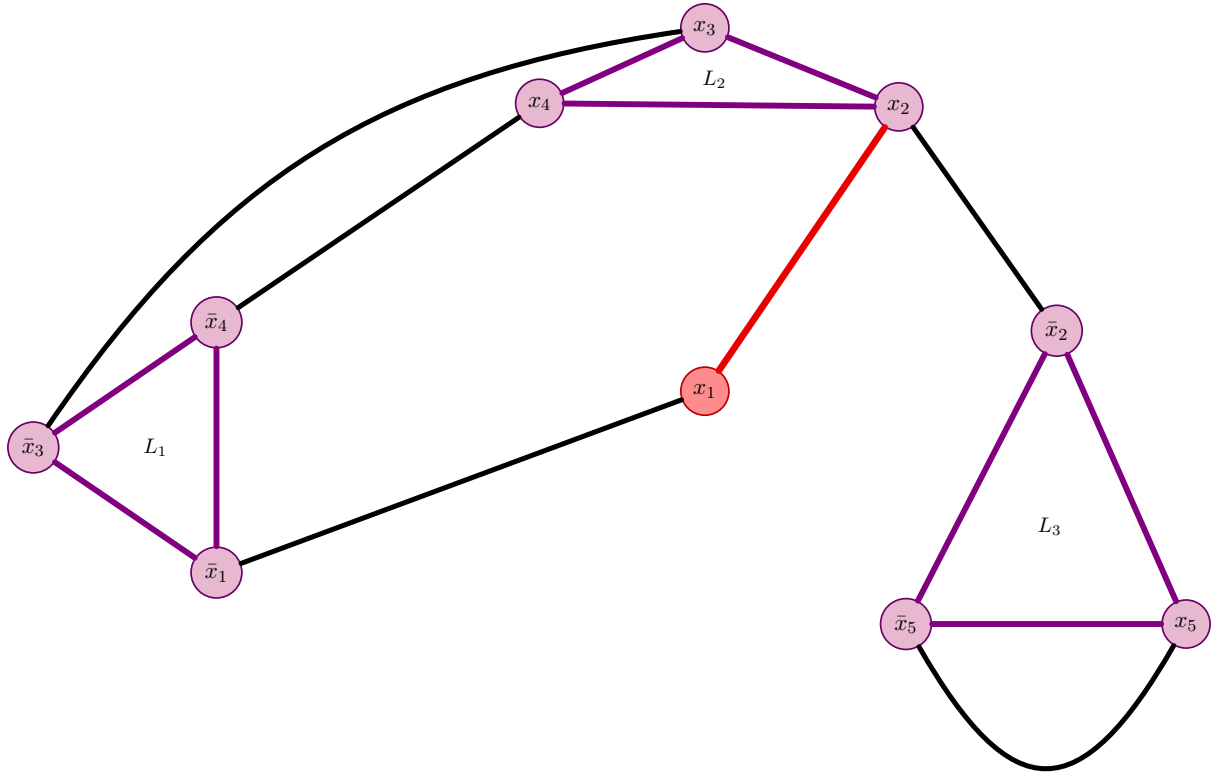
Distinguished indices: $\{4, 13, 58, 67, 72, 108, 126, 159, 166, 243, 256, 269\}$, 12 total.



Class 4 (VIII)

Representative of this ep-isomorphism class.

Distinguished indices: $\{7, 16, 56, 69, 75, 106, 128, 157, 168, 242, 255, 270\}$, 12 total.

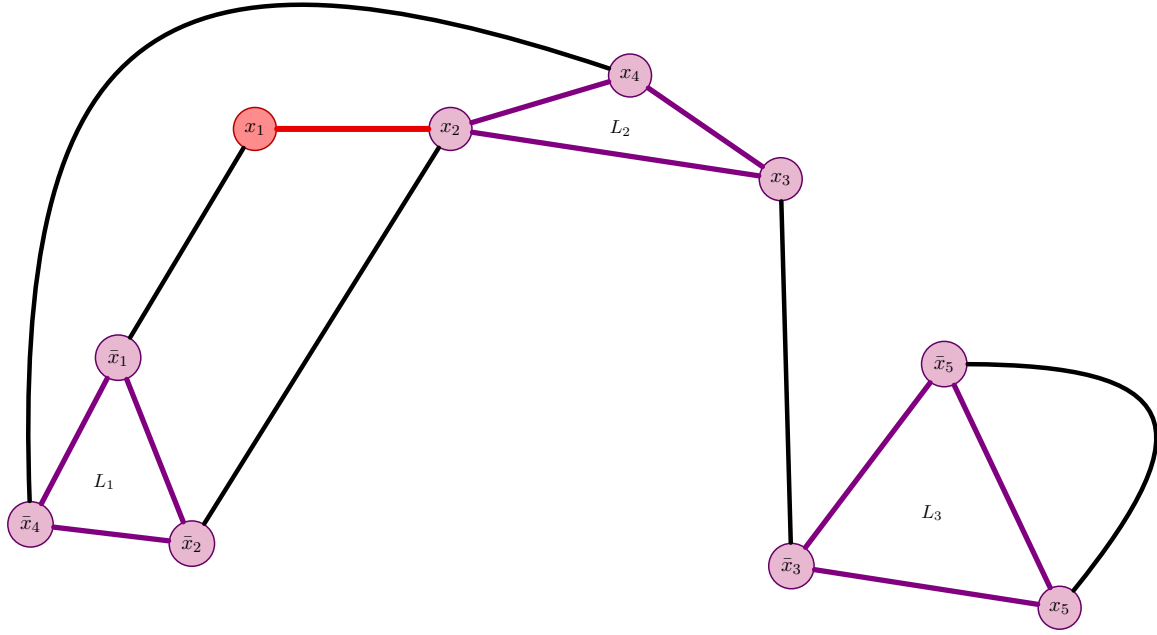


Class 5 (VI)

Representative of this ep-isomorphism class.

Distinguished indices:

$\{9, 10, 18, 20, 54, 55, 64, 70, 78, 79, 104, 105, 123, 130, 153, 155, 163, 170, 247, 248, 254, 258, 264, 267\}$,
24 total.

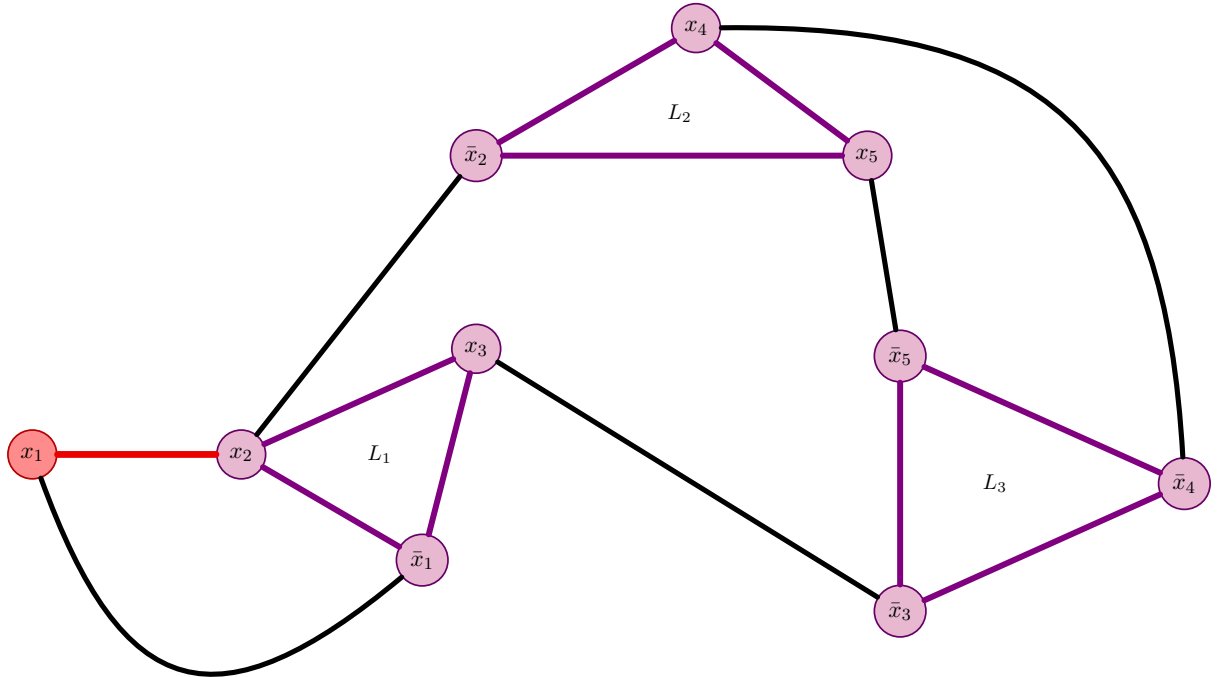


Class 6 (III)

Representative of this ep-isomorphism class.

Distinguished indices:

$\{21, 22, 27, 29, 81, 83, 87, 90, 131, 134, 136, 140, 181, 184, 186, 190, 191, 194, 195, 199, 201, 202, 207, 209\}$,
24 total.

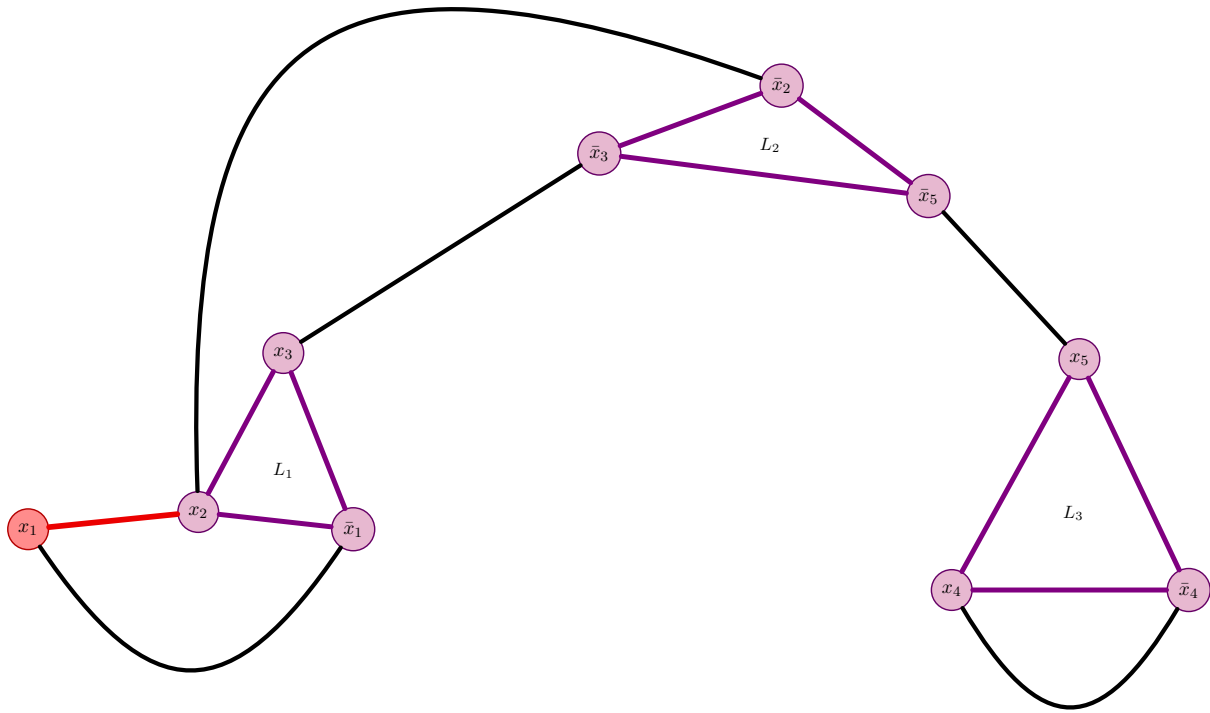


Class 7 (X)

Representative of this ep-isomorphism class.

Distinguished indices:

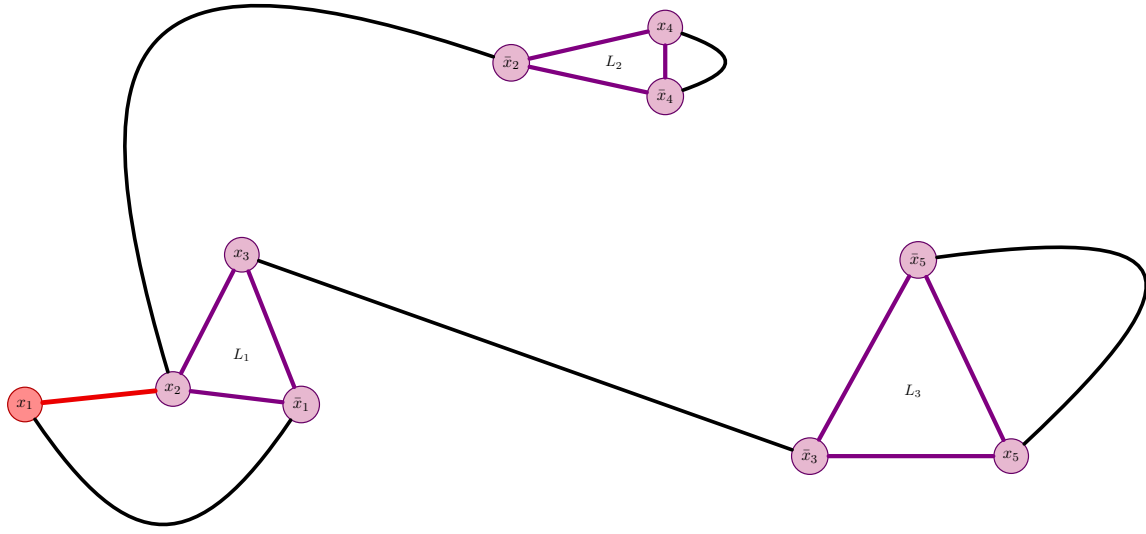
$\{23, 24, 25, 30, 82, 84, 86, 89, 132, 133, 137, 138, 182, 185, 188, 189, 192, 196, 197, 200, 203, 204, 205, 210\}$,
24 total.



Class 8 (XII)

Representative of this ep-isomorphism class.

Distinguished indices: $\{26, 28, 85, 88, 135, 139, 183, 187, 193, 198, 206, 208\}$, 12 total.

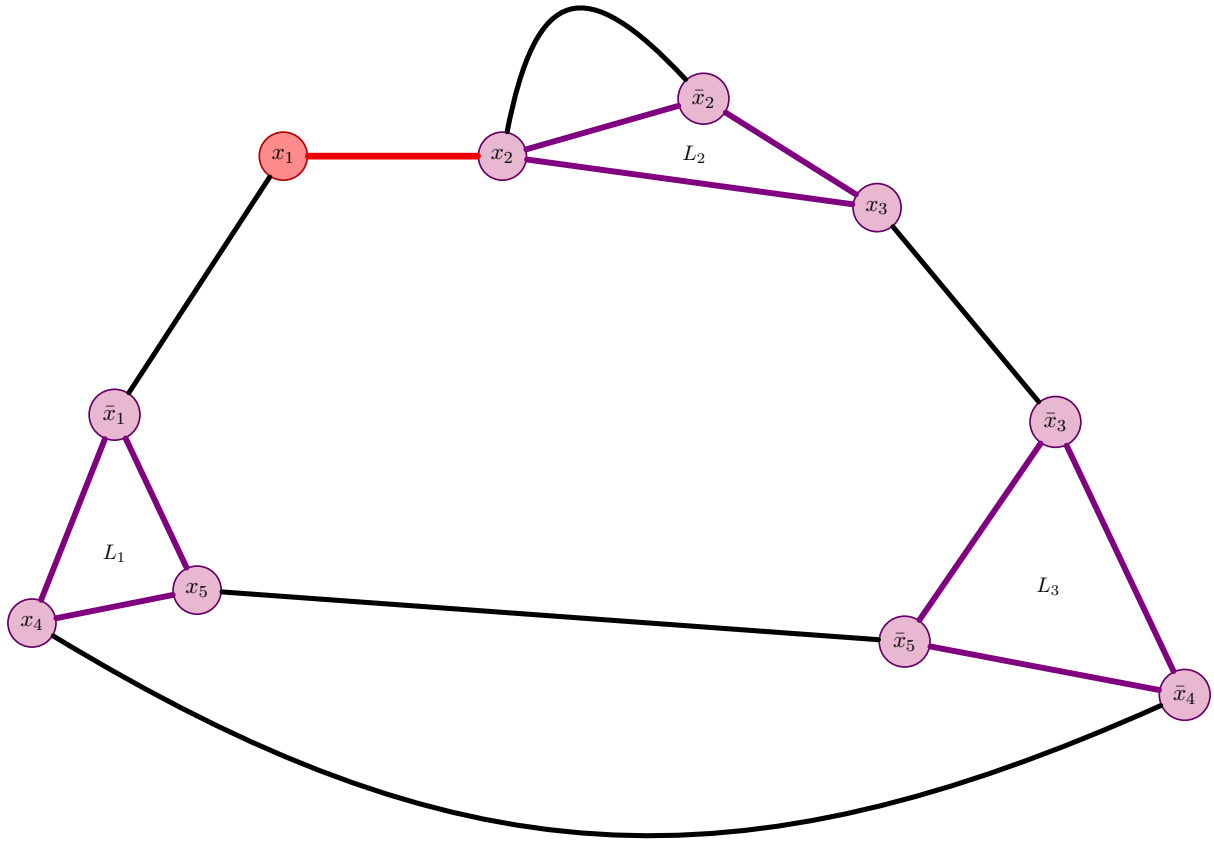


Class 9 (IV)

Representative of this ep-isomorphism class.

Distinguished indices:

$\{31, 32, 37, 39, 91, 93, 97, 100, 141, 144, 146, 150, 211, 214, 216, 220, 221, 224, 225, 229, 231, 232, 237, 239\}$,
24 total.

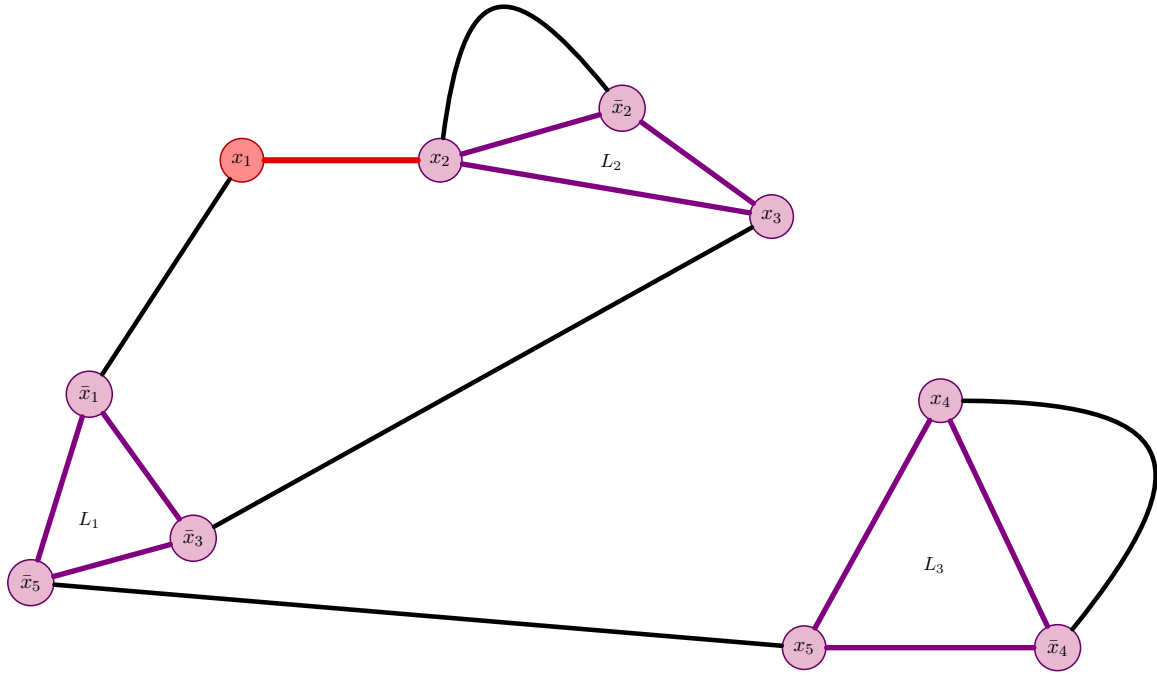


Class 10 (IX)

Representative of this ep-isomorphism class.

Distinguished indices:

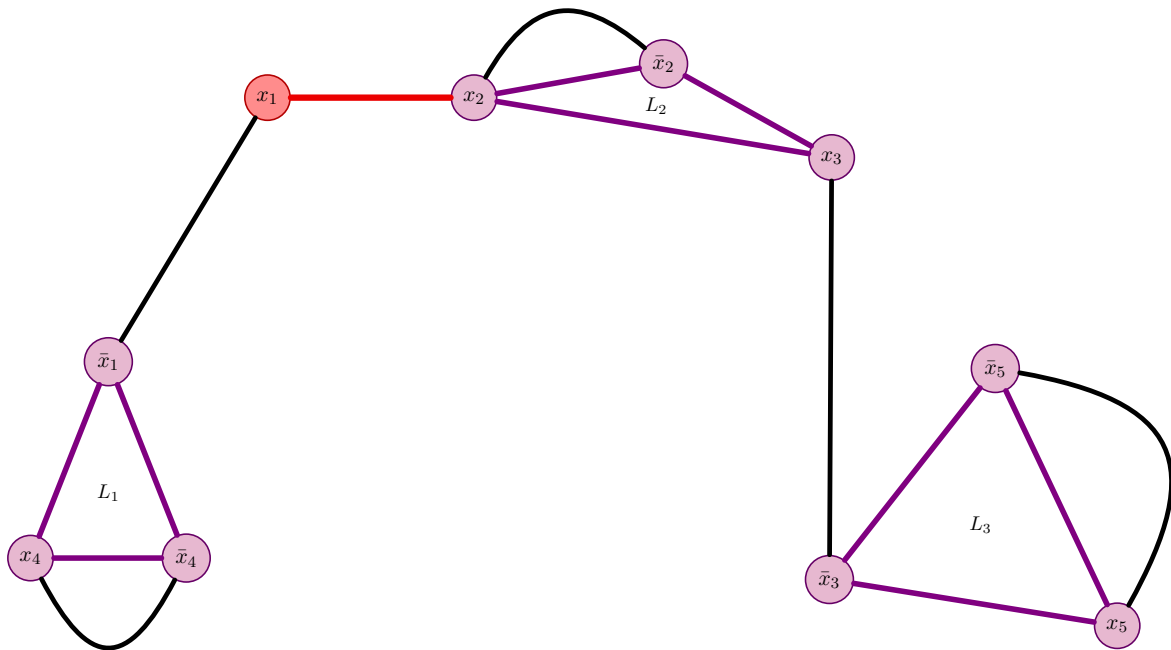
$\{33, 34, 35, 40, 92, 94, 96, 99, 142, 143, 147, 148, 212, 215, 218, 219, 222, 226, 227, 230, 233, 234, 235, 240\}$,
24 total.



Class 11 (XIII)

Representative of this ep-isomorphism class.

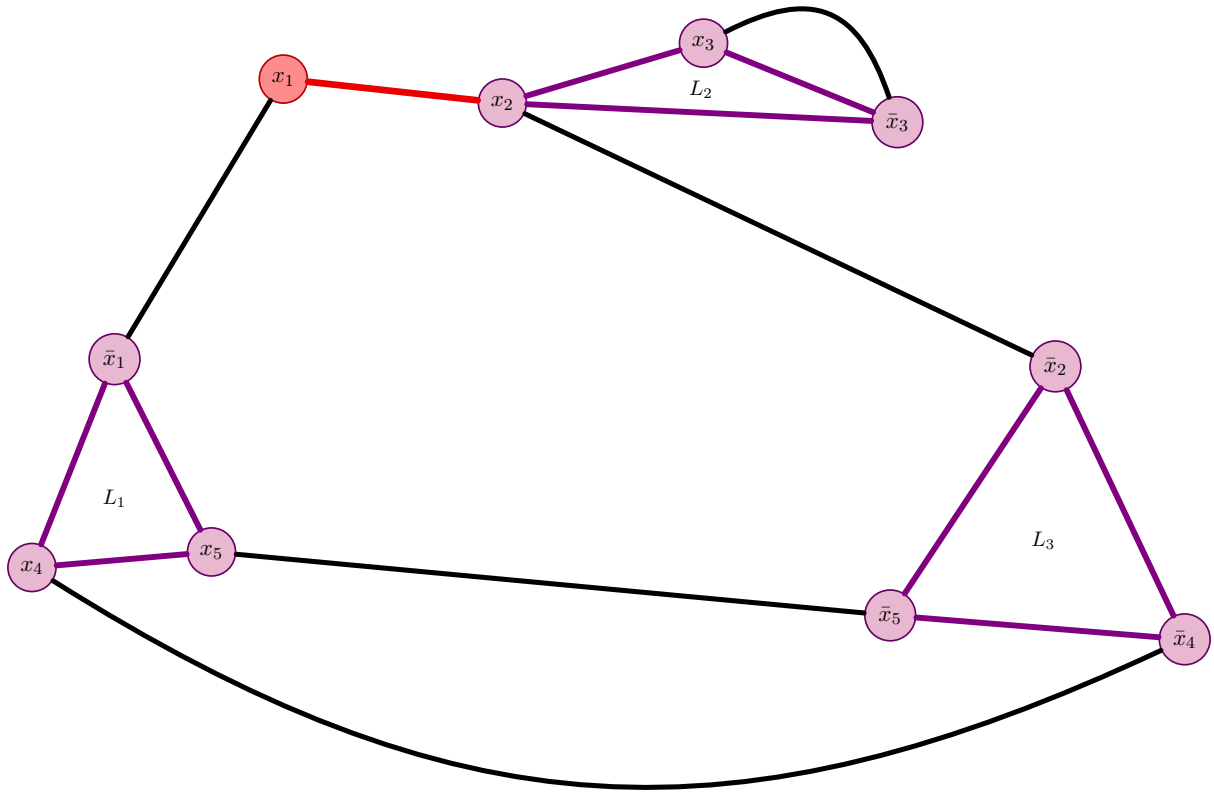
Distinguished indices: $\{36, 38, 95, 98, 145, 149, 213, 217, 223, 228, 236, 238\}$, 12 total.



Class 12 (V)

Representative of this ep-isomorphism class.

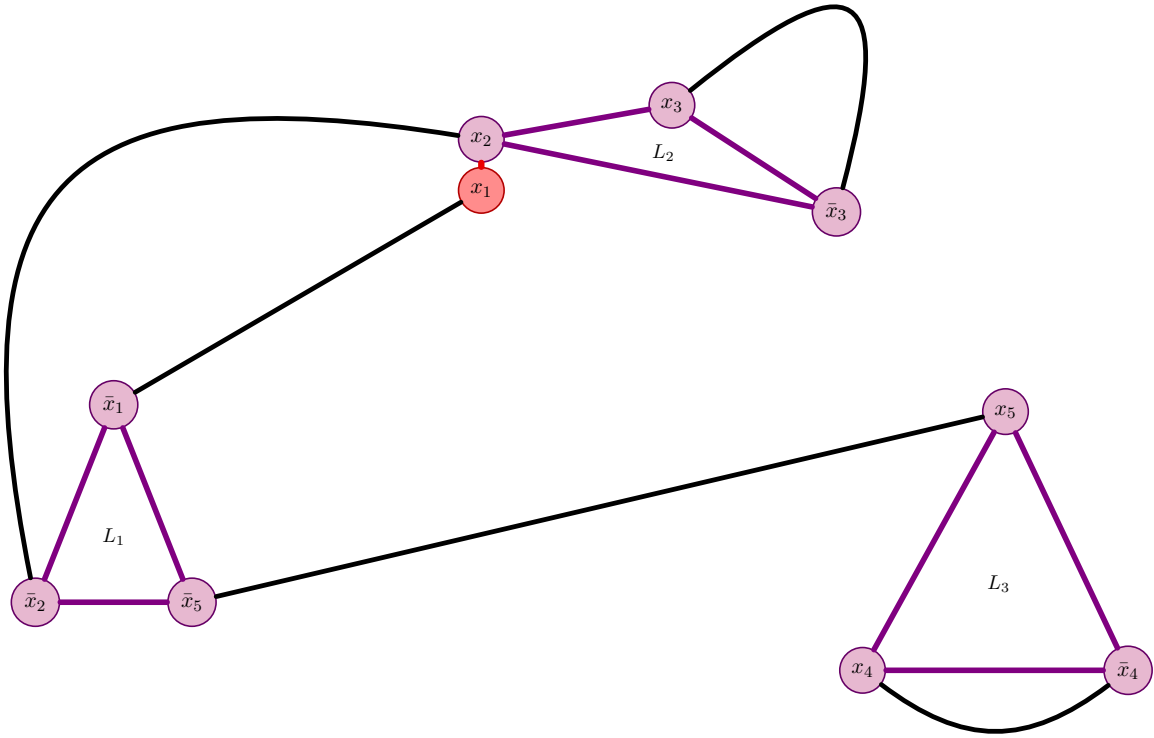
Distinguished indices: $\{41, 42, 47, 49, 111, 112, 117, 119, 171, 172, 177, 179\}$, 12 total.



Class 13 (XI)

Representative of this ep-isomorphism class.

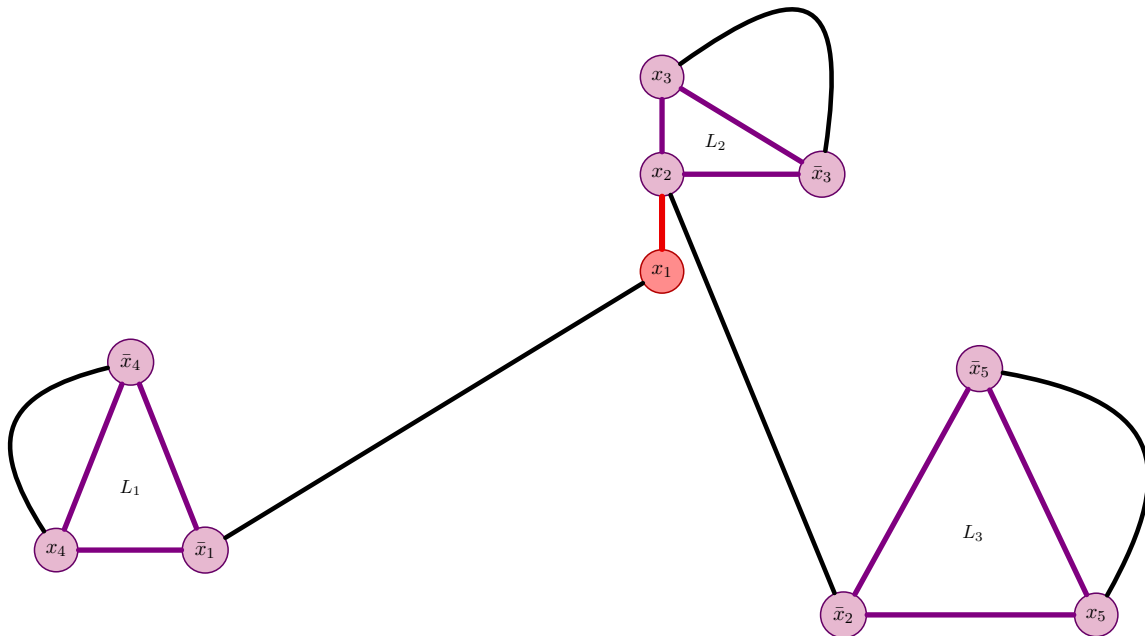
Distinguished indices: $\{43, 44, 45, 50, 113, 114, 115, 120, 173, 174, 175, 180\}$, 12 total.



Class 14 (XIV)

Representative of this ep-isomorphism class.

Distinguished indices: $\{46, 48, 116, 118, 176, 178\}$, 6 total.



B Output of Step 2 and Step 3 for the seed ltt structure (Tasks 1.3 and 1.4)

Regenerating this appendix. Run `python generate_step2_appendix_tex.py` and `python generate_step3_appendix_tex.py` from the repository root; they fill `step2_seed_appendix_generated.tex` and `step3_seed_appendix_generated.tex` with formatted L^AT_EX (matching `run_step2_seed.py` and `run_step3_seed.py`).

Seed *ltt* structure G

Red vertex

x_1 .

Red edge

$\{x_1, \bar{x}_5\}$.

Purple edges (9)

$P_1 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$

$P_2 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$

$P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$

Black edges (5)

$\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

Step 2 parameters

$d_r = x_1$, $d_p = \bar{x}_5$ (purple endpoint of the red edge).

Prompt alignment: rule (j;a) specifies red vertex d_r , red edge $\{d_r, \bar{d}_j\}$, and purple edges exactly as in G ; rule (j;b) adds purple edges from colored neighbors of d_j and inherits only purple edges not incident to d_j (see `ltt_structures.terminal_ga_step2` and `ltt_structures.terminal_gb_step2`). Directed-edge labels use (d_r, d_j) .

Membership in PLTT uses ep-isomorphism to one of the 270 distinguished graphs (same enumeration as `list_admissible_ltt_classes.py`).

Terminal graphs $\text{terminal}(G) = \{G_{1a}, G_{1b}, G_{2a}, G_{2b}\}$

1. G_{1a} : directed edge label $x_1 \rightarrow \bar{x}_3 x_1$, (rule 2a: $d_r \rightarrow d_j d_r$). **retained** (in PLTT).

Red vertex

x_1 .

Red edge

$\{x_1, x_3\}$.

Purple edges (9)

$P_1 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$

$P_2 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$

$P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$

Black edges (5)

$\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

2. G_{1b} : directed edge label $\bar{x}_3 \rightarrow x_1 \bar{x}_3$, (rule 2b: $d_j \rightarrow d_r d_j$). **retained** (in PLTT).

Red vertex

$\bar{x}_3$.

Red edge

$$\{\bar{x}_1, \bar{x}_3\}.$$

Purple edges (9)

$$P_1 : \{x_1, \bar{x}_4\}, \{x_1, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$$

$$P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$$

$$P_3 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$$

Black edges (5)

$$\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}.$$

3. G_{2a} : directed edge label $x_1 \rightarrow \bar{x}_4 x_1$, (rule 2a: $d_r \rightarrow d_j d_r$). **retained** (in PLTT).

Red vertex

$$x_1.$$

Red edge

$$\{x_1, x_4\}.$$

Purple edges (9)

$$P_1 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$$

$$P_2 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$$

$$P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$$

Black edges (5)

$$\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}.$$

4. G_{2b} : directed edge label $\bar{x}_4 \rightarrow x_1 \bar{x}_4$, (rule 2b: $d_j \rightarrow d_r d_j$). **retained** (in PLTT).

Red vertex

$$\bar{x}_4.$$

Red edge

$$\{\bar{x}_1, \bar{x}_4\}.$$

Purple edges (9)

$$P_1 : \{x_1, \bar{x}_3\}, \{x_1, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$$

$$P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$$

$$P_3 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$$

Black edges (5)

$$\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}.$$

Step 3 on seed output (Task 1.4)

Starting from Step 1 initialization $\text{GRAPHS} = \{G\}$, $\text{graphs}_1 = \{G\}$, $\text{boxes} = \{\text{graphs}_1\}$, $\text{GRAPHS}_1 = \{G\}$, apply Step 3 in the Step 2 order $\{G_{1a}, G_{1b}, G_{2a}, G_{2b}\}$.

1. G_{1a} : new ep-isomorphism class; add G_{1a} to GRAPHS and preAUT , create $\text{graphs}_2 = \{G_{1a}\}$, add G_{1a} to GRAPHS_2 , and add edge $G \rightarrow G_{1a}$ with label $x_1 \rightarrow \bar{x}_3 x_1$.

Red vertex

$$x_1.$$

Red edge

$$\{x_1, x_3\}.$$

Purple edges (9)

$$\begin{aligned} P_1 &: \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\} \\ P_2 &: \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\} \\ P_3 &: \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\} \end{aligned}$$

Black edges (5)

$$\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}.$$

2. G_{1b} : ep-isomorphic to G_{1a} (box graphs₂), not equal; add G_{1b} to GRAPHS and graphs₂, and add permutation edge $G_{1b} \xrightarrow{\varphi} G_{1a}$ with $\varphi = x_1 \mapsto \bar{x}_3, x_2 \mapsto x_2, x_3 \mapsto \bar{x}_1, x_4 \mapsto x_4, x_5 \mapsto x_5$.

Red vertex

$$\bar{x}_3.$$

Red edge

$$\{\bar{x}_1, \bar{x}_3\}.$$

Purple edges (9)

$$\begin{aligned} P_1 &: \{x_1, \bar{x}_4\}, \{x_1, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\} \\ P_2 &: \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\} \\ P_3 &: \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\} \end{aligned}$$

Black edges (5)

$$\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}.$$

3. G_{2a} : ep-isomorphic to G (box graphs₁), not equal; add G_{2a} to GRAPHS and graphs₁, and add permutation edge $G_{2a} \xrightarrow{\varphi} G$ with $\varphi = x_1 \mapsto x_1, x_2 \mapsto \bar{x}_3, x_3 \mapsto \bar{x}_2, x_4 \mapsto \bar{x}_5, x_5 \mapsto \bar{x}_4$.

Red vertex

$$x_1.$$

Red edge

$$\{x_1, x_4\}.$$

Purple edges (9)

$$\begin{aligned} P_1 &: \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\} \\ P_2 &: \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\} \\ P_3 &: \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\} \end{aligned}$$

Black edges (5)

$$\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}.$$

4. G_{2b} : ep-isomorphic to G (box graphs₁), not equal; add G_{2b} to GRAPHS and graphs₁, and add permutation edge $G_{2b} \xrightarrow{\varphi} G$ with $\varphi = x_1 \mapsto x_5, x_2 \mapsto x_3, x_3 \mapsto \bar{x}_4, x_4 \mapsto \bar{x}_1, x_5 \mapsto \bar{x}_2$.

Red vertex

$$\bar{x}_4.$$

Red edge

$$\{\bar{x}_1, \bar{x}_4\}.$$

Purple edges (9)

$$\begin{aligned} P_1 &: \{x_1, \bar{x}_3\}, \{x_1, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\} \\ P_2 &: \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\} \\ P_3 &: \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\} \end{aligned}$$

Black edges (5)

$$\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}.$$

Outcome after Step 3 for the seed. $\text{GRAPHS} = \{G, G_{1a}, G_{1b}, G_{2a}, G_{2b}\}$.
 $|\text{boxes}| = 2$, with:

- $\text{graphs}_1 = \{G, G_{2a}, G_{2b}\}$.
- $\text{graphs}_2 = \{G_{1a}, G_{1b}\}$.

$\text{GRAPHS}_2 = \{G_{1a}\}$.

preAUT edges added in this Step 3 pass.

- $G \xrightarrow{x_1 \mapsto \bar{x}_3, x_1} G_{1a}$.
- $G_{1b} \xrightarrow{\varphi} G_{1a}$ with $\varphi = x_1 \mapsto \bar{x}_3, x_2 \mapsto x_2, x_3 \mapsto \bar{x}_1, x_4 \mapsto x_4, x_5 \mapsto x_5$.
- $G_{2a} \xrightarrow{\varphi} G$ with $\varphi = x_1 \mapsto x_1, x_2 \mapsto \bar{x}_3, x_3 \mapsto \bar{x}_2, x_4 \mapsto \bar{x}_5, x_5 \mapsto \bar{x}_4$.
- $G_{2b} \xrightarrow{\varphi} G$ with $\varphi = x_1 \mapsto x_5, x_2 \mapsto x_3, x_3 \mapsto \bar{x}_4, x_4 \mapsto \bar{x}_1, x_5 \mapsto \bar{x}_2$.

C Output of preAUT and AUT (Section 1 algorithm)

Regenerating this appendix. Run `python generate_preaut_appendix_tex.py` from the repository root. This executes Steps 1–5 recursively from the seed until termination, checks Step 6 (ep-isomorphism-class coverage), and then computes Step 7 (AUT as the disjoint union of maximal strongly connected components).

Global outcome

Total nodes in preAUT (after omitting components with no edges): 42. Total directed edges in preAUT: 78. Total boxes (ep-isomorphism classes used by Step 3): 14. Maximal strongly connected components kept in AUT (box-collapsed): 1. Nodes in AUT (box-collapsed): 14. Directed edges in AUT (box-collapsed): 39.

Levels GRAPHS_k

- $\text{GRAPHS}_1 = \{G\}$.
- $\text{GRAPHS}_2 = \{G_1\}$.
- $\text{GRAPHS}_3 = \{G_5, G_6\}$.
- $\text{GRAPHS}_4 = \{G_{10}\}$.
- $\text{GRAPHS}_5 = \{G_{12}\}$.
- $\text{GRAPHS}_6 = \{G_{16}, G_{17}\}$.
- $\text{GRAPHS}_7 = \{G_{18}, G_{19}, G_{20}\}$.
- $\text{GRAPHS}_8 = \{G_{30}, G_{31}\}$.
- $\text{GRAPHS}_9 = \{G_{32}\}$.

Boxes

- graphs_1 (Class 1, Box II) = $\{G_5, G_{26}, G_{29}\}$.
- graphs_2 (Class 2, Box I) = $\{G, G_2, G_4, G_7, G_8, G_9\}$.
- graphs_3 (Class 3, Box VII) = $\{G_{18}, G_{22}, G_{23}\}$.
- graphs_4 (Class 4, Box VIII) = $\{G_{10}, G_{11}\}$.
- graphs_5 (Class 5, Box VI) = $\{G_{16}, G_{34}, G_{35}\}$.
- graphs_6 (Class 6, Box III) = $\{G_1, G_3\}$.
- graphs_7 (Class 7, Box X) = $\{G_{12}, G_{13}, G_{14}, G_{15}\}$.
- graphs_8 (Class 8, Box XII) = $\{G_{20}, G_{21}\}$.
- graphs_9 (Class 9, Box IV) = $\{G_6, G_{27}, G_{28}\}$.
- graphs_{10} (Class 10, Box IX) = $\{G_{17}, G_{36}, G_{37}\}$.
- graphs_{11} (Class 11, Box XIII) = $\{G_{30}, G_{38}, G_{39}\}$.
- graphs_{12} (Class 12, Box V) = $\{G_{19}, G_{24}, G_{25}\}$.
- graphs_{13} (Class 13, Box XI) = $\{G_{31}, G_{40}, G_{41}\}$.
- graphs_{14} (Class 14, Box XIV) = $\{G_{32}, G_{33}\}$.

Directed edges of preAUT

1. $G \xrightarrow{x_1 \rightarrow \bar{x}_3} G_1$.
2. $G \xrightarrow{x_1 \rightarrow \bar{x}_4} G_2$.
3. $G_2 \xrightarrow{\varphi} G$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto \bar{x}_3, x_3 \mapsto \bar{x}_2, x_4 \mapsto \bar{x}_5, x_5 \mapsto \bar{x}_4$.
4. $G \xrightarrow{\bar{x}_3 \rightarrow x_1} G_3$.
5. $G_3 \xrightarrow{\varphi} G_1$, where $\varphi = x_1 \mapsto \bar{x}_3, x_2 \mapsto x_2, x_3 \mapsto \bar{x}_1, x_4 \mapsto x_4, x_5 \mapsto x_5$.
6. $G \xrightarrow{\bar{x}_4 \rightarrow x_1} G_4$.
7. $G_4 \xrightarrow{\varphi} G$, where $\varphi = x_1 \mapsto x_5, x_2 \mapsto x_3, x_3 \mapsto \bar{x}_4, x_4 \mapsto \bar{x}_1, x_5 \mapsto \bar{x}_2$.
8. $G_1 \xrightarrow{x_1 \rightarrow \bar{x}_2} G_5$.
9. $G_1 \xrightarrow{\bar{x}_2 \rightarrow x_1} G_6$.
10. $G_5 \xrightarrow{x_1 \rightarrow x_4} G_7$.
11. $G_7 \xrightarrow{\varphi} G$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto x_3, x_4 \mapsto x_5, x_5 \mapsto x_4$.
12. $G_5 \xrightarrow{x_1 \rightarrow x_5} G$.
13. $G_5 \xrightarrow{x_4 \rightarrow x_1} G_8$.
14. $G_8 \xrightarrow{\varphi} G$, where $\varphi = x_1 \mapsto x_5, x_2 \mapsto x_4, x_3 \mapsto \bar{x}_3, x_4 \mapsto x_1, x_5 \mapsto x_2$.
15. $G_5 \xrightarrow{x_5 \rightarrow x_1} G_9$.

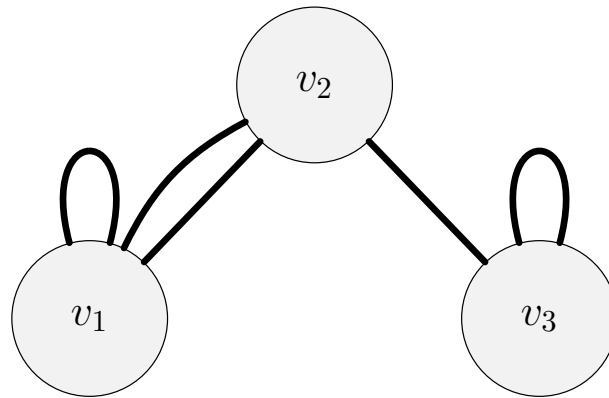
16. $G_9 \xrightarrow{\varphi} G$, where $\varphi = x_1 \mapsto x_5, x_2 \mapsto x_4, x_3 \mapsto \bar{x}_3, x_4 \mapsto x_2, x_5 \mapsto x_1$.
17. $G_6 \xrightarrow{x_1 \mapsto \bar{x}_2, x_1} G_5$.
18. $G_6 \xrightarrow{x_3 \mapsto \bar{x}_2, x_3} G_{10}$.
19. $G_6 \xrightarrow{\bar{x}_2 \mapsto x_1, \bar{x}_2} G_6$.
20. $G_6 \xrightarrow{\bar{x}_2 \mapsto x_3, \bar{x}_2} G_{11}$.
21. $G_{11} \xrightarrow{\varphi} G_{10}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto \bar{x}_3, x_3 \mapsto \bar{x}_2, x_4 \mapsto \bar{x}_4, x_5 \mapsto \bar{x}_5$.
22. $G_{10} \xrightarrow{x_3 \mapsto x_4, x_3} G_{12}$.
23. $G_{10} \xrightarrow{x_3 \mapsto x_5, x_3} G_{13}$.
24. $G_{13} \xrightarrow{\varphi} G_{12}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto x_3, x_4 \mapsto x_5, x_5 \mapsto x_4$.
25. $G_{10} \xrightarrow{x_4 \mapsto x_3, x_4} G_{14}$.
26. $G_{14} \xrightarrow{\varphi} G_{12}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto x_4, x_4 \mapsto x_3, x_5 \mapsto x_5$.
27. $G_{10} \xrightarrow{x_5 \mapsto x_3, x_5} G_{15}$.
28. $G_{15} \xrightarrow{\varphi} G_{12}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto x_4, x_4 \mapsto x_5, x_5 \mapsto x_3$.
29. $G_{12} \xrightarrow{x_3 \mapsto \bar{x}_5, x_3} G_{16}$.
30. $G_{12} \xrightarrow{\bar{x}_5 \mapsto x_3, \bar{x}_5} G_{17}$.
31. $G_{16} \xrightarrow{x_2 \mapsto x_3, x_2} G_{18}$.
32. $G_{16} \xrightarrow{x_3 \mapsto x_2, x_3} G_{19}$.
33. $G_{16} \xrightarrow{x_3 \mapsto x_4, x_3} G_{12}$.
34. $G_{16} \xrightarrow{x_4 \mapsto x_3, x_4} G_{14}$.
35. $G_{17} \xrightarrow{x_3 \mapsto \bar{x}_5, x_3} G_{16}$.
36. $G_{17} \xrightarrow{\bar{x}_4 \mapsto \bar{x}_5, \bar{x}_4} G_{20}$.
37. $G_{17} \xrightarrow{\bar{x}_5 \mapsto \bar{x}_4, \bar{x}_5} G_{21}$.
38. $G_{21} \xrightarrow{\varphi} G_{20}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto x_3, x_4 \mapsto x_5, x_5 \mapsto x_4$.
39. $G_{17} \xrightarrow{\bar{x}_5 \mapsto x_3, \bar{x}_5} G_{17}$.
40. $G_{18} \xrightarrow{x_2 \mapsto \bar{x}_4, x_2} G_{22}$.
41. $G_{22} \xrightarrow{\varphi} G_{18}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto \bar{x}_4, x_4 \mapsto \bar{x}_3, x_5 \mapsto \bar{x}_5$.
42. $G_{18} \xrightarrow{x_2 \mapsto \bar{x}_5, x_2} G_{23}$.
43. $G_{23} \xrightarrow{\varphi} G_{18}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto \bar{x}_4, x_4 \mapsto \bar{x}_5, x_5 \mapsto \bar{x}_3$.
44. $G_{18} \xrightarrow{\bar{x}_4 \mapsto x_2, \bar{x}_4} G_{24}$.
45. $G_{24} \xrightarrow{\varphi} G_{19}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto \bar{x}_4, x_4 \mapsto \bar{x}_3, x_5 \mapsto \bar{x}_5$.
46. $G_{18} \xrightarrow{\bar{x}_5 \mapsto x_2, \bar{x}_5} G_{25}$.

47. $G_{25} \xrightarrow{\varphi} G_{19}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto \bar{x}_4, x_4 \mapsto \bar{x}_5, x_5 \mapsto \bar{x}_3$.
48. $G_{19} \xrightarrow{x_1 \rightarrow x_3 x_1} G_{26}$.
49. $G_{26} \xrightarrow{\varphi} G_5$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto \bar{x}_3, x_3 \mapsto \bar{x}_2, x_4 \mapsto \bar{x}_4, x_5 \mapsto \bar{x}_5$.
50. $G_{19} \xrightarrow{x_3 \rightarrow \bar{x}_1 x_3} G_{27}$.
51. $G_{27} \xrightarrow{\varphi} G_6$, where $\varphi = x_1 \mapsto \bar{x}_1, x_2 \mapsto \bar{x}_3, x_3 \mapsto \bar{x}_2, x_4 \mapsto \bar{x}_4, x_5 \mapsto \bar{x}_5$.
52. $G_{19} \xrightarrow{x_3 \rightarrow x_1 x_3} G_{28}$.
53. $G_{28} \xrightarrow{\varphi} G_6$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto \bar{x}_3, x_3 \mapsto \bar{x}_2, x_4 \mapsto \bar{x}_4, x_5 \mapsto \bar{x}_5$.
54. $G_{19} \xrightarrow{\bar{x}_1 \rightarrow x_3 \bar{x}_1} G_{29}$.
55. $G_{29} \xrightarrow{\varphi} G_5$, where $\varphi = x_1 \mapsto \bar{x}_1, x_2 \mapsto \bar{x}_3, x_3 \mapsto \bar{x}_2, x_4 \mapsto \bar{x}_4, x_5 \mapsto \bar{x}_5$.
56. $G_{20} \xrightarrow{x_2 \rightarrow \bar{x}_4 x_2} G_{30}$.
57. $G_{20} \xrightarrow{\bar{x}_4 \rightarrow x_2 \bar{x}_4} G_{31}$.
58. $G_{30} \xrightarrow{x_2 \rightarrow \bar{x}_4 x_2} G_{30}$.
59. $G_{30} \xrightarrow{x_2 \rightarrow x_5 x_2} G_{32}$.
60. $G_{30} \xrightarrow{x_5 \rightarrow x_2 x_5} G_{33}$.
61. $G_{33} \xrightarrow{\varphi} G_{32}$, where $\varphi = x_1 \mapsto x_3, x_2 \mapsto x_5, x_3 \mapsto x_1, x_4 \mapsto x_4, x_5 \mapsto x_2$.
62. $G_{30} \xrightarrow{\bar{x}_4 \rightarrow x_2 \bar{x}_4} G_{31}$.
63. $G_{31} \xrightarrow{x_1 \rightarrow \bar{x}_4 x_1} G_{34}$.
64. $G_{34} \xrightarrow{\varphi} G_{16}$, where $\varphi = x_1 \mapsto x_3, x_2 \mapsto x_4, x_3 \mapsto x_1, x_4 \mapsto x_5, x_5 \mapsto x_2$.
65. $G_{31} \xrightarrow{\bar{x}_1 \rightarrow \bar{x}_4 \bar{x}_1} G_{35}$.
66. $G_{35} \xrightarrow{\varphi} G_{16}$, where $\varphi = x_1 \mapsto \bar{x}_3, x_2 \mapsto x_4, x_3 \mapsto x_1, x_4 \mapsto x_5, x_5 \mapsto x_2$.
67. $G_{31} \xrightarrow{\bar{x}_4 \rightarrow \bar{x}_1 \bar{x}_4} G_{36}$.
68. $G_{36} \xrightarrow{\varphi} G_{17}$, where $\varphi = x_1 \mapsto \bar{x}_3, x_2 \mapsto x_4, x_3 \mapsto x_1, x_4 \mapsto x_5, x_5 \mapsto x_2$.
69. $G_{31} \xrightarrow{\bar{x}_4 \rightarrow x_1 \bar{x}_4} G_{37}$.
70. $G_{37} \xrightarrow{\varphi} G_{17}$, where $\varphi = x_1 \mapsto x_3, x_2 \mapsto x_4, x_3 \mapsto x_1, x_4 \mapsto x_5, x_5 \mapsto x_2$.
71. $G_{32} \xrightarrow{x_2 \rightarrow \bar{x}_3 x_2} G_{38}$.
72. $G_{38} \xrightarrow{\varphi} G_{30}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto x_4, x_4 \mapsto x_3, x_5 \mapsto \bar{x}_5$.
73. $G_{32} \xrightarrow{x_2 \rightarrow x_3 x_2} G_{39}$.
74. $G_{39} \xrightarrow{\varphi} G_{30}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto \bar{x}_4, x_4 \mapsto x_3, x_5 \mapsto \bar{x}_5$.
75. $G_{32} \xrightarrow{x_3 \rightarrow x_2 x_3} G_{40}$.
76. $G_{40} \xrightarrow{\varphi} G_{31}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto \bar{x}_4, x_4 \mapsto x_3, x_5 \mapsto \bar{x}_5$.
77. $G_{32} \xrightarrow{\bar{x}_3 \rightarrow x_2 \bar{x}_3} G_{41}$.
78. $G_{41} \xrightarrow{\varphi} G_{31}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto x_4, x_4 \mapsto x_3, x_5 \mapsto \bar{x}_5$.

Underlying graph isomorphism classes (unlabeled)

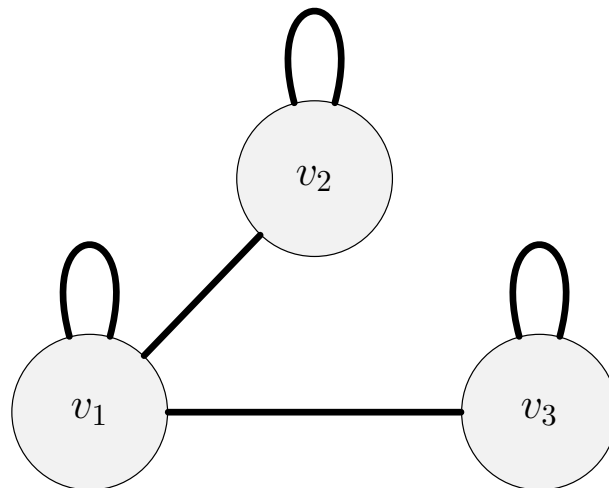
Collapsing each colored component (purple cliques and red edge attachment) yields an underlying multigraph. Up to standard graph isomorphism, the preAUT/AUT dictionary graphs realize the following classes.

1. U_1 .



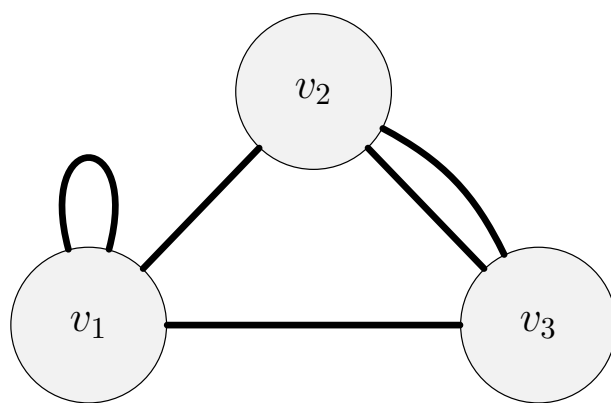
U_1

2. U_2 .



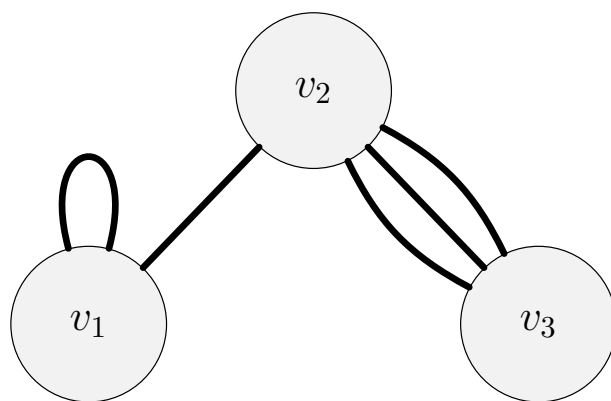
U_2

3. U_3 .



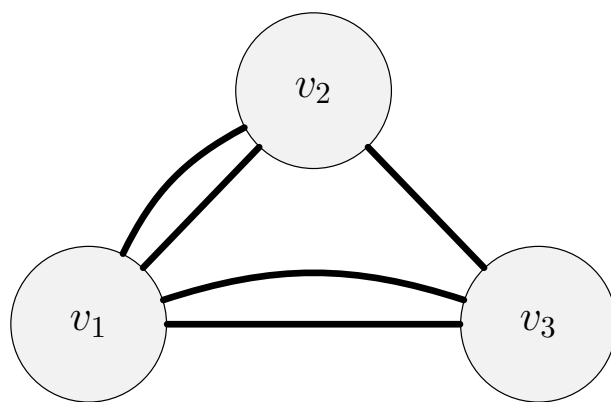
U_3

4. U_4 .



U_4

5. U_5 .



U_5

Dictionary of graph labels used in preAUT/AUT

For each graph label used above, this table records the corresponding ltt structure data.

1. G :

- underlying graph class = U_5 .
- red vertex = x_1 .
- red edge = $\{x_1, \bar{x}_5\}$.
- purple edges ($|\mathcal{P}| = 9$):
 - $P_1 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 - $P_2 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 - $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

2. G_1 :

- underlying graph class = U_3 .
- red vertex = x_1 .
- red edge = $\{x_1, x_3\}$.
- purple edges ($|\mathcal{P}| = 9$):
 - $P_1 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 - $P_2 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 - $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

3. G_2 :

- underlying graph class = U_5 .
- red vertex = x_1 .
- red edge = $\{x_1, x_4\}$.
- purple edges ($|\mathcal{P}| = 9$):
 - $P_1 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 - $P_2 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 - $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

4. G_3 :

- underlying graph class = U_3 .
- red vertex = $\bar{x}_3$.
- red edge = $\{\bar{x}_1, \bar{x}_3\}$.
- purple edges ($|\mathcal{P}| = 9$):
 - $P_1 : \{x_1, \bar{x}_4\}, \{x_1, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$
 - $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 - $P_3 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

5. G_4 :

- underlying graph class = U_5 .

- red vertex = $\bar{x}_4$.
- red edge = $\{\bar{x}_1, \bar{x}_4\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_3\}, \{x_1, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

6. G_5 :

- underlying graph class = U_5 .
- red vertex = x_1 .
- red edge = $\{x_1, x_2\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_2 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

7. G_6 :

- underlying graph class = U_3 .
- red vertex = $\bar{x}_2$.
- red edge = $\{\bar{x}_1, \bar{x}_2\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, x_3\}, \{x_1, \bar{x}_1\}, \{x_3, \bar{x}_1\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

8. G_7 :

- underlying graph class = U_5 .
- red vertex = x_1 .
- red edge = $\{x_1, \bar{x}_4\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_2 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

9. G_8 :

- underlying graph class = U_5 .
- red vertex = x_4 .
- red edge = $\{x_4, \bar{x}_1\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, x_2\}, \{x_1, x_5\}, \{x_2, x_5\}$
 $P_2 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.

- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

10. G_9 :

- underlying graph class = U_5 .
- red vertex = x_5 .
- red edge = $\{x_5, \bar{x}_1\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, x_2\}, \{x_1, x_4\}, \{x_2, x_4\}$
 $P_2 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

11. G_{10} :

- underlying graph class = U_4 .
- red vertex = x_3 .
- red edge = $\{x_2, x_3\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

12. G_{11} :

- underlying graph class = U_4 .
- red vertex = $\bar{x}_2$.
- red edge = $\{\bar{x}_2, \bar{x}_3\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, x_3\}, \{x_1, \bar{x}_1\}, \{x_3, \bar{x}_1\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

13. G_{12} :

- underlying graph class = U_1 .
- red vertex = x_3 .
- red edge = $\{x_3, \bar{x}_4\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

14. G_{13} :

- underlying graph class = U_1 .
- red vertex = x_3 .

- red edge = $\{x_3, \bar{x}_5\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

15. G_{14} :

- underlying graph class = U_1 .
- red vertex = x_4 .
- red edge = $\{x_4, \bar{x}_3\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_3\}, \{x_2, x_5\}, \{x_3, x_5\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

16. G_{15} :

- underlying graph class = U_1 .
- red vertex = x_5 .
- red edge = $\{x_5, \bar{x}_3\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_3\}, \{x_2, x_4\}, \{x_3, x_4\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

17. G_{16} :

- underlying graph class = U_4 .
- red vertex = x_3 .
- red edge = $\{x_3, x_5\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

18. G_{17} :

- underlying graph class = U_1 .
- red vertex = $\bar{x}_5$.
- red edge = $\{\bar{x}_3, \bar{x}_5\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_4\}$.

- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

19. G_{18} :

- underlying graph class = U_4 .
- red vertex = x_2 .
- red edge = $\{x_2, \bar{x}_3\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_3, x_4\}, \{x_3, x_5\}, \{x_4, x_5\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

20. G_{19} :

- underlying graph class = U_3 .
- red vertex = x_3 .
- red edge = $\{x_3, \bar{x}_2\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

21. G_{20} :

- underlying graph class = U_2 .
- red vertex = $\bar{x}_4$.
- red edge = $\{x_5, \bar{x}_4\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

22. G_{21} :

- underlying graph class = U_2 .
- red vertex = $\bar{x}_5$.
- red edge = $\{x_4, \bar{x}_5\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_4\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

23. G_{22} :

- underlying graph class = U_4 .
- red vertex = x_2 .

- red edge = $\{x_2, x_4\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_3, x_4\}, \{x_3, x_5\}, \{x_4, x_5\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

24. G_{23} :

- underlying graph class = U_4 .
- red vertex = x_2 .
- red edge = $\{x_2, x_5\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_3, x_4\}, \{x_3, x_5\}, \{x_4, x_5\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

25. G_{24} :

- underlying graph class = U_3 .
- red vertex = $\bar{x}_4$.
- red edge = $\{\bar{x}_2, \bar{x}_4\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, \bar{x}_3\}, \{x_2, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$
 $P_3 : \{x_3, x_4\}, \{x_3, x_5\}, \{x_4, x_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

26. G_{25} :

- underlying graph class = U_3 .
- red vertex = $\bar{x}_5$.
- red edge = $\{\bar{x}_2, \bar{x}_5\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, \bar{x}_3\}, \{x_2, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_4\}$
 $P_3 : \{x_3, x_4\}, \{x_3, x_5\}, \{x_4, x_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

27. G_{26} :

- underlying graph class = U_5 .
- red vertex = x_1 .
- red edge = $\{x_1, \bar{x}_3\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_2 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.

- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

28. G_{27} :

- underlying graph class = U_3 .
- red vertex = x_3 .
- red edge = $\{x_1, x_3\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

29. G_{28} :

- underlying graph class = U_3 .
- red vertex = x_3 .
- red edge = $\{x_3, \bar{x}_1\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

30. G_{29} :

- underlying graph class = U_5 .
- red vertex = $\bar{x}_1$.
- red edge = $\{\bar{x}_1, \bar{x}_3\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, x_3\}, \{x_1, \bar{x}_2\}, \{x_3, \bar{x}_2\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

31. G_{30} :

- underlying graph class = U_2 .
- red vertex = x_2 .
- red edge = $\{x_2, x_4\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$
 $P_3 : \{x_4, x_5\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_4\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

32. G_{31} :

- underlying graph class = U_1 .
- red vertex = $\bar{x}_4$.

- red edge = $\{\bar{x}_2, \bar{x}_4\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

33. G_{32} :

- underlying graph class = U_2 .
- red vertex = x_2 .
- red edge = $\{x_2, \bar{x}_5\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$
 $P_3 : \{x_4, x_5\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_4\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

34. G_{33} :

- underlying graph class = U_2 .
- red vertex = x_5 .
- red edge = $\{x_5, \bar{x}_2\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_4\}, \{x_2, \bar{x}_4\}, \{x_4, \bar{x}_4\}$
 $P_3 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

35. G_{34} :

- underlying graph class = U_4 .
- red vertex = x_1 .
- red edge = $\{x_1, x_4\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_2 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$
 $P_3 : \{\bar{x}_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_4\}, \{\bar{x}_2, \bar{x}_4\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

36. G_{35} :

- underlying graph class = U_4 .
- red vertex = $\bar{x}_1$.
- red edge = $\{x_4, \bar{x}_1\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_2\}, \{x_1, \bar{x}_4\}, \{\bar{x}_2, \bar{x}_4\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$.

- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

37. G_{36} :

- underlying graph class = U_1 .
- red vertex = $\bar{x}_4$.
- red edge = $\{x_1, \bar{x}_4\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

38. G_{37} :

- underlying graph class = U_1 .
- red vertex = $\bar{x}_4$.
- red edge = $\{\bar{x}_1, \bar{x}_4\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

39. G_{38} :

- underlying graph class = U_2 .
- red vertex = x_2 .
- red edge = $\{x_2, x_3\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$
 $P_3 : \{x_4, x_5\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_4\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

40. G_{39} :

- underlying graph class = U_2 .
- red vertex = x_2 .
- red edge = $\{x_2, \bar{x}_3\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$
 $P_3 : \{x_4, x_5\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_4\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

41. G_{40} :

- underlying graph class = U_1 .
- red vertex = x_3 .

- red edge = $\{x_3, \bar{x}_2\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, \bar{x}_3\}, \{x_2, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$
 $P_3 : \{x_4, x_5\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_4\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

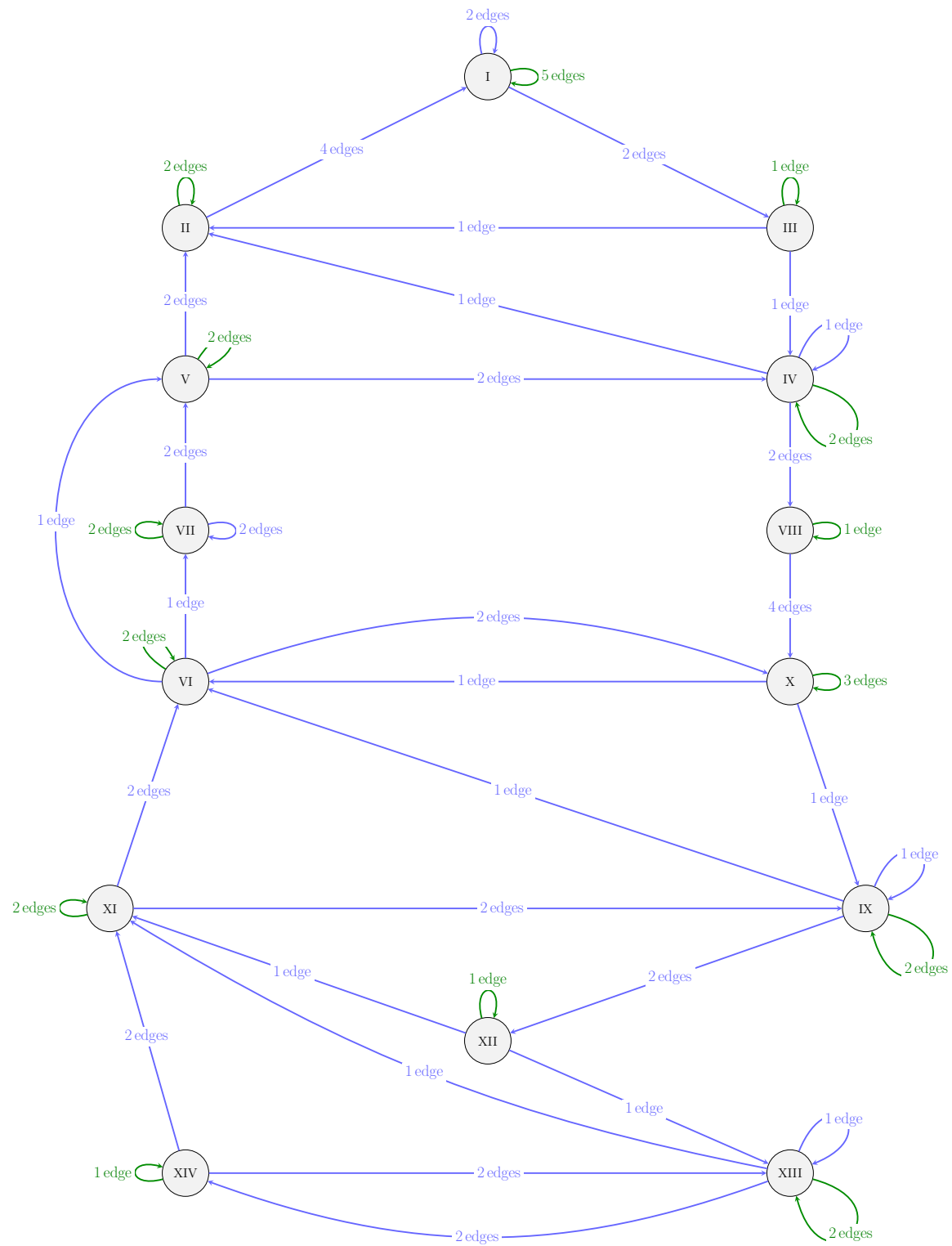
42. G_{41} :

- underlying graph class = U_1 .
- red vertex = $\bar{x}_3$.
- red edge = $\{\bar{x}_2, \bar{x}_3\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_3\}, \{x_2, \bar{x}_5\}, \{x_3, \bar{x}_5\}$
 $P_3 : \{x_4, x_5\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_4\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

Visual depiction of AUT by component (box-collapsed)

AUT is computed from the box-collapsed preAUT (one node per box). Retained AUT components are listed sequentially. Nodes are labeled by Roman box IDs (I–XIV). Blue (Step 2) edge labels count only non-permutation arrows along that directed box pair; green labels count permutation (φ) edges, so blue and green labels sum to the total parallel preAUT arrows between those boxes.

AUT component SCC_1 .



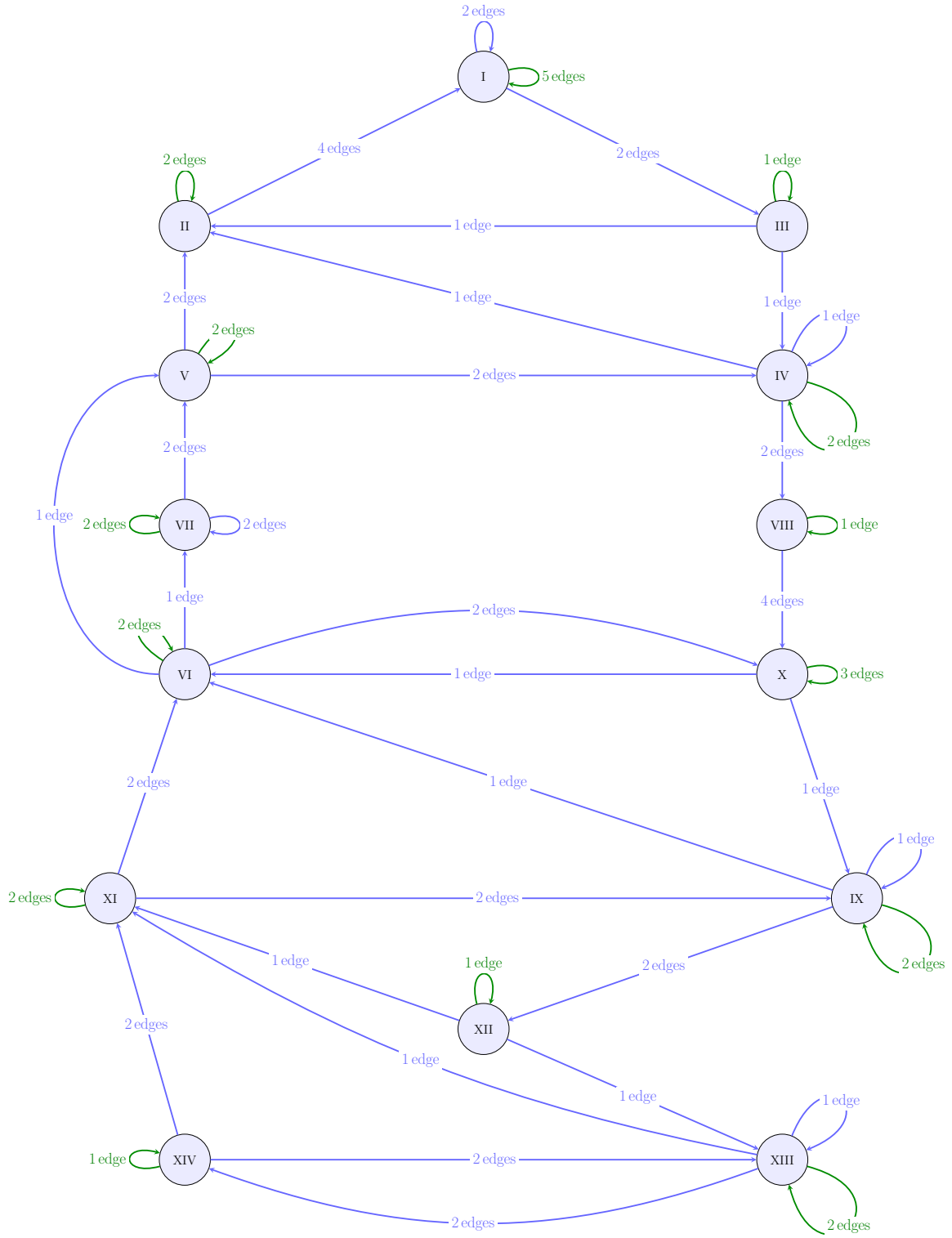
Node index list for this component.

- Box II = Class₁ (II).
- Box I = Class₂ (I).
- Box VII = Class₃ (VII).
- Box VIII = Class₄ (VIII).
- Box VI = Class₅ (VI).
- Box III = Class₆ (III).
- Box X = Class₇ (X).
- Box XII = Class₈ (XII).
- Box IV = Class₉ (IV).
- Box IX = Class₁₀ (IX).
- Box XIII = Class₁₁ (XIII).
- Box V = Class₁₂ (V).
- Box XI = Class₁₃ (XI).
- Box XIV = Class₁₄ (XIV).

Visual depiction of full preAUT (one node per box)

This is the box-collapsed preAUT: each node is one Roman box (I–XIV). A directed edge is drawn from one box to another whenever preAUT has at least one edge between their corresponding classes. As in the AUT diagrams, blue counts Step 2 transitions only and green counts permutation (φ) edges along each directed box pair.

Box-collapsed preAUT component 1.



Box index list for this preAUT diagram.

- Box II = $\text{graphs}_1 = \{G_{26}, G_{29}, G_5\}$.
- Box I = $\text{graphs}_2 = \{G, G_2, G_4, G_7, G_8, G_9\}$.
- Box VII = $\text{graphs}_3 = \{G_{18}, G_{22}, G_{23}\}$.
- Box VIII = $\text{graphs}_4 = \{G_{10}, G_{11}\}$.

- Box VI = $\text{graphs}_5 = \{G_{16}, G_{34}, G_{35}\}$.
- Box III = $\text{graphs}_6 = \{G_1, G_3\}$.
- Box X = $\text{graphs}_7 = \{G_{12}, G_{13}, G_{14}, G_{15}\}$.
- Box XII = $\text{graphs}_8 = \{G_{20}, G_{21}\}$.
- Box IV = $\text{graphs}_9 = \{G_{27}, G_{28}, G_6\}$.
- Box IX = $\text{graphs}_{10} = \{G_{17}, G_{36}, G_{37}\}$.
- Box XIII = $\text{graphs}_{11} = \{G_{30}, G_{38}, G_{39}\}$.
- Box V = $\text{graphs}_{12} = \{G_{19}, G_{24}, G_{25}\}$.
- Box XI = $\text{graphs}_{13} = \{G_{31}, G_{40}, G_{41}\}$.
- Box XIV = $\text{graphs}_{14} = \{G_{32}, G_{33}\}$.

preAUT edges omitted in AUT (cross-SCC/exiting edges)

The AUT figures keep only edges whose source and target are in the same maximal SCC. The following preAUT edges leave those SCCs and are therefore omitted from AUT.

1. From SCC_1 : none.

Visual depiction of omitted preAUT edges by SCC

For each AUT component SCC_i , we draw the edges that start in SCC_i and leave it in preAUT. Internal AUT edges are omitted in these diagrams.

Omitted preAUT edges from SCC_1 . None.

Step 6: coverage check across ep-isomorphism classes

Covered ep-isomorphism classes: 14 out of 14. All classes are represented by the constructed preAUT, so no additional seed run is required.

Step 7: maximal SCCs and AUT (box-collapsed)

Each maximal strongly connected component (SCC) of the box-collapsed preAUT is listed below; AUT is their disjoint union.

SCC decomposition of preAUT.

1. $\text{SCC}_1 = \{\text{Class}_1 \text{ (II)}, \text{Class}_{10} \text{ (IX)}, \text{Class}_{11} \text{ (XIII)}, \text{Class}_{12} \text{ (V)}, \text{Class}_{13} \text{ (XI)}, \text{Class}_{14} \text{ (XIV)}, \text{Class}_2 \text{ (I)}, \text{Class}_3 \text{ (V)}\}$

Directed edges of AUT. These are exactly the box-collapsed preAUT edges whose end-points lie in the same maximal SCC.

1. $\text{Class}_1 \text{ (II)} \xrightarrow{2 \text{ edges}} \text{Class}_1 \text{ (II)}$ (inside SCC_1).
2. $\text{Class}_1 \text{ (II)} \xrightarrow{4 \text{ edges}} \text{Class}_2 \text{ (I)}$ (inside SCC_1).
3. $\text{Class}_2 \text{ (I)} \xrightarrow{7 \text{ edges}} \text{Class}_2 \text{ (I)}$ (inside SCC_1).

4. $\text{Class}_2 \text{ (I)} \xrightarrow{2 \text{ edges}} \text{Class}_6 \text{ (III)} \text{ (inside } \text{SCC}_1 \text{)}.$
5. $\text{Class}_3 \text{ (VII)} \xrightarrow{4 \text{ edges}} \text{Class}_3 \text{ (VII)} \text{ (inside } \text{SCC}_1 \text{)}.$
6. $\text{Class}_3 \text{ (VII)} \xrightarrow{2 \text{ edges}} \text{Class}_{12} \text{ (V)} \text{ (inside } \text{SCC}_1 \text{)}.$
7. $\text{Class}_4 \text{ (VIII)} \xrightarrow{1 \text{ edge}} \text{Class}_4 \text{ (VIII)} \text{ (inside } \text{SCC}_1 \text{)}.$
8. $\text{Class}_4 \text{ (VIII)} \xrightarrow{4 \text{ edges}} \text{Class}_7 \text{ (X)} \text{ (inside } \text{SCC}_1 \text{)}.$
9. $\text{Class}_5 \text{ (VI)} \xrightarrow{1 \text{ edge}} \text{Class}_3 \text{ (VII)} \text{ (inside } \text{SCC}_1 \text{)}.$
10. $\text{Class}_5 \text{ (VI)} \xrightarrow{2 \text{ edges}} \text{Class}_5 \text{ (VI)} \text{ (inside } \text{SCC}_1 \text{)}.$
11. $\text{Class}_5 \text{ (VI)} \xrightarrow{2 \text{ edges}} \text{Class}_7 \text{ (X)} \text{ (inside } \text{SCC}_1 \text{)}.$
12. $\text{Class}_5 \text{ (VI)} \xrightarrow{1 \text{ edge}} \text{Class}_{12} \text{ (V)} \text{ (inside } \text{SCC}_1 \text{)}.$
13. $\text{Class}_6 \text{ (III)} \xrightarrow{1 \text{ edge}} \text{Class}_1 \text{ (II)} \text{ (inside } \text{SCC}_1 \text{)}.$
14. $\text{Class}_6 \text{ (III)} \xrightarrow{1 \text{ edge}} \text{Class}_6 \text{ (III)} \text{ (inside } \text{SCC}_1 \text{)}.$
15. $\text{Class}_6 \text{ (III)} \xrightarrow{1 \text{ edge}} \text{Class}_9 \text{ (IV)} \text{ (inside } \text{SCC}_1 \text{)}.$
16. $\text{Class}_7 \text{ (X)} \xrightarrow{1 \text{ edge}} \text{Class}_5 \text{ (VI)} \text{ (inside } \text{SCC}_1 \text{)}.$
17. $\text{Class}_7 \text{ (X)} \xrightarrow{3 \text{ edges}} \text{Class}_7 \text{ (X)} \text{ (inside } \text{SCC}_1 \text{)}.$
18. $\text{Class}_7 \text{ (X)} \xrightarrow{1 \text{ edge}} \text{Class}_{10} \text{ (IX)} \text{ (inside } \text{SCC}_1 \text{)}.$
19. $\text{Class}_8 \text{ (XII)} \xrightarrow{1 \text{ edge}} \text{Class}_8 \text{ (XII)} \text{ (inside } \text{SCC}_1 \text{)}.$
20. $\text{Class}_8 \text{ (XII)} \xrightarrow{1 \text{ edge}} \text{Class}_{11} \text{ (XIII)} \text{ (inside } \text{SCC}_1 \text{)}.$
21. $\text{Class}_8 \text{ (XII)} \xrightarrow{1 \text{ edge}} \text{Class}_{13} \text{ (XI)} \text{ (inside } \text{SCC}_1 \text{)}.$
22. $\text{Class}_9 \text{ (IV)} \xrightarrow{1 \text{ edge}} \text{Class}_1 \text{ (II)} \text{ (inside } \text{SCC}_1 \text{)}.$
23. $\text{Class}_9 \text{ (IV)} \xrightarrow{2 \text{ edges}} \text{Class}_4 \text{ (VIII)} \text{ (inside } \text{SCC}_1 \text{)}.$
24. $\text{Class}_9 \text{ (IV)} \xrightarrow{3 \text{ edges}} \text{Class}_9 \text{ (IV)} \text{ (inside } \text{SCC}_1 \text{)}.$
25. $\text{Class}_{10} \text{ (IX)} \xrightarrow{1 \text{ edge}} \text{Class}_5 \text{ (VI)} \text{ (inside } \text{SCC}_1 \text{)}.$
26. $\text{Class}_{10} \text{ (IX)} \xrightarrow{2 \text{ edges}} \text{Class}_8 \text{ (XII)} \text{ (inside } \text{SCC}_1 \text{)}.$
27. $\text{Class}_{10} \text{ (IX)} \xrightarrow{3 \text{ edges}} \text{Class}_{10} \text{ (IX)} \text{ (inside } \text{SCC}_1 \text{)}.$
28. $\text{Class}_{11} \text{ (XIII)} \xrightarrow{3 \text{ edges}} \text{Class}_{11} \text{ (XIII)} \text{ (inside } \text{SCC}_1 \text{)}.$
29. $\text{Class}_{11} \text{ (XIII)} \xrightarrow{1 \text{ edge}} \text{Class}_{13} \text{ (XI)} \text{ (inside } \text{SCC}_1 \text{)}.$
30. $\text{Class}_{11} \text{ (XIII)} \xrightarrow{2 \text{ edges}} \text{Class}_{14} \text{ (XIV)} \text{ (inside } \text{SCC}_1 \text{)}.$

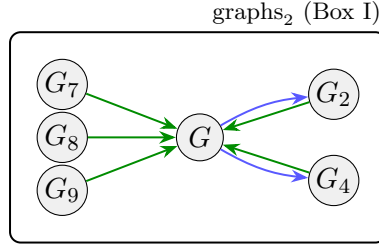
31. $\text{Class}_{12}(\text{V}) \xrightarrow{2 \text{ edges}} \text{Class}_1(\text{II})$ (inside SCC_1).
32. $\text{Class}_{12}(\text{V}) \xrightarrow{2 \text{ edges}} \text{Class}_9(\text{IV})$ (inside SCC_1).
33. $\text{Class}_{12}(\text{V}) \xrightarrow{2 \text{ edges}} \text{Class}_{12}(\text{V})$ (inside SCC_1).
34. $\text{Class}_{13}(\text{XI}) \xrightarrow{2 \text{ edges}} \text{Class}_5(\text{VI})$ (inside SCC_1).
35. $\text{Class}_{13}(\text{XI}) \xrightarrow{2 \text{ edges}} \text{Class}_{10}(\text{IX})$ (inside SCC_1).
36. $\text{Class}_{13}(\text{XI}) \xrightarrow{2 \text{ edges}} \text{Class}_{13}(\text{XI})$ (inside SCC_1).
37. $\text{Class}_{14}(\text{XIV}) \xrightarrow{2 \text{ edges}} \text{Class}_{11}(\text{XIII})$ (inside SCC_1).
38. $\text{Class}_{14}(\text{XIV}) \xrightarrow{2 \text{ edges}} \text{Class}_{13}(\text{XI})$ (inside SCC_1).
39. $\text{Class}_{14}(\text{XIV}) \xrightarrow{1 \text{ edge}} \text{Class}_{14}(\text{XIV})$ (inside SCC_1).

D Internal preAUT maps within each ep-isomorphism class

The Roman numeral for each class matches the translation table in the next section. Each diagram shows only edges whose source and target lie in the same class (edges induced inside one graphs_k box).

Regeneration. Run `python generate_box_internal_maps_appendix_tex.py` from the repository root.

Class 2 (Box I): internal preAUT maps



Graph data (lft structures in this box).

1. G :

- underlying graph class = U_5 .
- red vertex = x_1 .
- red edge = $\{x_1, \bar{x}_5\}$.
- purple edges ($|\mathcal{P}| = 9$):
 - $P_1 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 - $P_2 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 - $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

2. G_2 :

- underlying graph class = U_5 .

- red vertex = x_1 .
- red edge = $\{x_1, x_4\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_2 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

3. G_4 :

- underlying graph class = U_5 .
- red vertex = $\bar{x}_4$.
- red edge = $\{\bar{x}_1, \bar{x}_4\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_3\}, \{x_1, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

4. G_7 :

- underlying graph class = U_5 .
- red vertex = x_1 .
- red edge = $\{x_1, \bar{x}_4\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_2 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

5. G_8 :

- underlying graph class = U_5 .
- red vertex = x_4 .
- red edge = $\{x_4, \bar{x}_1\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, x_2\}, \{x_1, x_5\}, \{x_2, x_5\}$
 $P_2 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

6. G_9 :

- underlying graph class = U_5 .
- red vertex = x_5 .
- red edge = $\{x_5, \bar{x}_1\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, x_2\}, \{x_1, x_4\}, \{x_2, x_4\}$
 $P_2 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

Arrows. Blue: Step 2 edge label. Green: permutation φ from ep-isomorphism (Step 3).

Maps represented (internal to Box I).

1. $G \xrightarrow{x_1 \rightarrow \bar{x}_4 x_1} G_2$ (Step 2).
2. $G \xrightarrow{\bar{x}_4 \rightarrow x_1 \bar{x}_4} G_4$ (Step 2).
3. $G_2 \xrightarrow{\varphi} G$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto \bar{x}_3, x_3 \mapsto \bar{x}_2, x_4 \mapsto \bar{x}_5, x_5 \mapsto \bar{x}_4$ (permutation / ep-isomorphism).
4. $G_4 \xrightarrow{\varphi} G$, where $\varphi = x_1 \mapsto x_5, x_2 \mapsto x_3, x_3 \mapsto \bar{x}_4, x_4 \mapsto \bar{x}_1, x_5 \mapsto \bar{x}_2$ (permutation / ep-isomorphism).
5. $G_7 \xrightarrow{\varphi} G$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto x_3, x_4 \mapsto x_5, x_5 \mapsto x_4$ (permutation / ep-isomorphism).
6. $G_8 \xrightarrow{\varphi} G$, where $\varphi = x_1 \mapsto x_5, x_2 \mapsto x_4, x_3 \mapsto \bar{x}_3, x_4 \mapsto x_1, x_5 \mapsto x_2$ (permutation / ep-isomorphism).
7. $G_9 \xrightarrow{\varphi} G$, where $\varphi = x_1 \mapsto x_5, x_2 \mapsto x_4, x_3 \mapsto \bar{x}_3, x_4 \mapsto x_2, x_5 \mapsto x_1$ (permutation / ep-isomorphism).

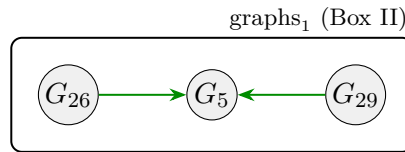
Incoming preAUT edges (into this box).

1. $G_5 \xrightarrow{x_1 \rightarrow x_5 x_1} G$ (graphs₁, Box II) (Step 2).
2. $G_5 \xrightarrow{x_1 \rightarrow x_4 x_1} G_7$ (graphs₁, Box II) (Step 2).
3. $G_5 \xrightarrow{x_4 \rightarrow x_1 x_4} G_8$ (graphs₁, Box II) (Step 2).
4. $G_5 \xrightarrow{x_5 \rightarrow x_1 x_5} G_9$ (graphs₁, Box II) (Step 2).

Outgoing preAUT edges (from this box).

1. $G \xrightarrow{x_1 \rightarrow \bar{x}_3 x_1} G_1$ (graphs₆, Box III) (Step 2).
2. $G \xrightarrow{\bar{x}_3 \rightarrow x_1 \bar{x}_3} G_3$ (graphs₆, Box III) (Step 2).

Class 1 (Box II): internal preAUT maps



Graph data (labeled structures in this box).

1. G_5 :
 - underlying graph class = U_5 .
 - red vertex = x_1 .
 - red edge = $\{x_1, x_2\}$.
 - purple edges ($|\mathcal{P}| = 9$):
 - $P_1 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 - $P_2 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 - $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.

- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

2. G_{26} :

- underlying graph class = U_5 .
- red vertex = x_1 .
- red edge = $\{x_1, \bar{x}_3\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_2 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

3. G_{29} :

- underlying graph class = U_5 .
- red vertex = $\bar{x}_1$.
- red edge = $\{\bar{x}_1, \bar{x}_3\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, x_3\}, \{x_1, \bar{x}_2\}, \{x_3, \bar{x}_2\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

Arrows. Blue: Step 2 edge label. Green: permutation φ from ep-isomorphism (Step 3).

Maps represented (internal to Box II).

1. $G_{26} \xrightarrow{\varphi} G_5$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto \bar{x}_3, x_3 \mapsto \bar{x}_2, x_4 \mapsto \bar{x}_4, x_5 \mapsto \bar{x}_5$ (permutation / ep-isomorphism).
2. $G_{29} \xrightarrow{\varphi} G_5$, where $\varphi = x_1 \mapsto \bar{x}_1, x_2 \mapsto \bar{x}_3, x_3 \mapsto \bar{x}_2, x_4 \mapsto \bar{x}_4, x_5 \mapsto \bar{x}_5$ (permutation / ep-isomorphism).

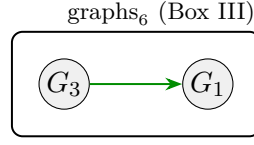
Incoming preAUT edges (into this box).

1. $G_1 \xrightarrow{x_1 \rightarrow \bar{x}_2 x_1} G_5$ (graphs₆, Box III) (Step 2).
2. $G_{19} \xrightarrow{x_1 \rightarrow x_3 x_1} G_{26}$ (graphs₁₂, Box V) (Step 2).
3. $G_{19} \xrightarrow{\bar{x}_1 \rightarrow x_3 \bar{x}_1} G_{29}$ (graphs₁₂, Box V) (Step 2).
4. $G_6 \xrightarrow{x_1 \rightarrow \bar{x}_2 x_1} G_5$ (graphs₉, Box IV) (Step 2).

Outgoing preAUT edges (from this box).

1. $G_5 \xrightarrow{x_1 \rightarrow x_5 x_1} G$ (graphs₂, Box I) (Step 2).
2. $G_5 \xrightarrow{x_1 \rightarrow x_4 x_1} G_7$ (graphs₂, Box I) (Step 2).
3. $G_5 \xrightarrow{x_4 \rightarrow x_1 x_4} G_8$ (graphs₂, Box I) (Step 2).
4. $G_5 \xrightarrow{x_5 \rightarrow x_1 x_5} G_9$ (graphs₂, Box I) (Step 2).

Class 6 (Box III): internal preAUT maps



Graph data (lft structures in this box).

1. G_1 :

- underlying graph class = U_3 .
- red vertex = x_1 .
- red edge = $\{x_1, x_3\}$.
- purple edges ($|\mathcal{P}| = 9$):
 - $P_1 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 - $P_2 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 - $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

2. G_3 :

- underlying graph class = U_3 .
- red vertex = $\bar{x}_3$.
- red edge = $\{\bar{x}_1, \bar{x}_3\}$.
- purple edges ($|\mathcal{P}| = 9$):
 - $P_1 : \{x_1, \bar{x}_4\}, \{x_1, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$
 - $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 - $P_3 : \{x_3, \bar{x}_1\}, \{x_3, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

Arrows. Blue: Step 2 edge label. Green: permutation φ from ep-isomorphism (Step 3).

Maps represented (internal to Box III).

1. $G_3 \xrightarrow{\varphi} G_1$, where $\varphi = x_1 \mapsto \bar{x}_3, x_2 \mapsto x_2, x_3 \mapsto \bar{x}_1, x_4 \mapsto x_4, x_5 \mapsto x_5$ (*permutation / ep-isomorphism*).

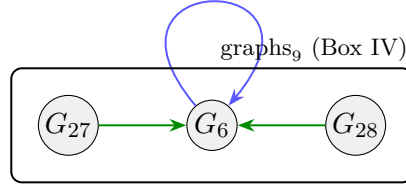
Incoming preAUT edges (into this box).

1. $G \xrightarrow{x_1 \rightarrow \bar{x}_3 \ x_1} G_1$ (graphs₂, Box I) (*Step 2*).
2. $G \xrightarrow{\bar{x}_3 \rightarrow x_1 \ \bar{x}_3} G_3$ (graphs₂, Box I) (*Step 2*).

Outgoing preAUT edges (from this box).

1. $G_1 \xrightarrow{x_1 \rightarrow \bar{x}_2 \ x_1} G_5$ (graphs₁, Box II) (*Step 2*).
2. $G_1 \xrightarrow{\bar{x}_2 \rightarrow x_1 \ \bar{x}_2} G_6$ (graphs₉, Box IV) (*Step 2*).

Class 9 (Box IV): internal preAUT maps



Graph data (lft structures in this box).

1. G_6 :

- underlying graph class = U_3 .
- red vertex = $\bar{x}_2$.
- red edge = $\{\bar{x}_1, \bar{x}_2\}$.
- purple edges ($|\mathcal{P}| = 9$):
 - $P_1 : \{x_1, x_3\}, \{x_1, \bar{x}_1\}, \{x_3, \bar{x}_1\}$
 - $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 - $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

2. G_{27} :

- underlying graph class = U_3 .
- red vertex = x_3 .
- red edge = $\{x_1, x_3\}$.
- purple edges ($|\mathcal{P}| = 9$):
 - $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 - $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 - $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

3. G_{28} :

- underlying graph class = U_3 .
- red vertex = x_3 .
- red edge = $\{x_3, \bar{x}_1\}$.
- purple edges ($|\mathcal{P}| = 9$):
 - $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 - $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 - $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

Arrows. Blue: Step 2 edge label. Green: permutation φ from ep-isomorphism (Step 3).

Maps represented (internal to Box IV).

1. $G_{27} \xrightarrow{\varphi} G_6$, where $\varphi = x_1 \mapsto \bar{x}_1, x_2 \mapsto \bar{x}_3, x_3 \mapsto \bar{x}_2, x_4 \mapsto \bar{x}_4, x_5 \mapsto \bar{x}_5$ (permutation / ep-isomorphism).
2. $G_{28} \xrightarrow{\varphi} G_6$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto \bar{x}_3, x_3 \mapsto \bar{x}_2, x_4 \mapsto \bar{x}_4, x_5 \mapsto \bar{x}_5$ (permutation / ep-isomorphism).

3. $G_6 \xrightarrow{\bar{x}_2 \rightarrow x_1 \bar{x}_2} G_6$ (*Step 2*).

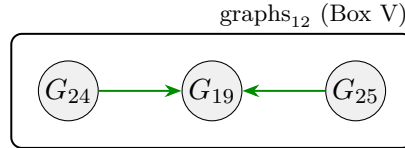
Incoming preAUT edges (into this box).

1. $G_1 \xrightarrow{\bar{x}_2 \rightarrow x_1 \bar{x}_2} G_6$ (graphs₆, Box III) (*Step 2*).
2. $G_{19} \xrightarrow{x_3 \rightarrow \bar{x}_1 x_3} G_{27}$ (graphs₁₂, Box V) (*Step 2*).
3. $G_{19} \xrightarrow{x_3 \rightarrow x_1 x_3} G_{28}$ (graphs₁₂, Box V) (*Step 2*).

Outgoing preAUT edges (from this box).

1. $G_6 \xrightarrow{x_3 \rightarrow \bar{x}_2 x_3} G_{10}$ (graphs₄, Box VIII) (*Step 2*).
2. $G_6 \xrightarrow{\bar{x}_2 \rightarrow x_3 \bar{x}_2} G_{11}$ (graphs₄, Box VIII) (*Step 2*).
3. $G_6 \xrightarrow{x_1 \rightarrow \bar{x}_2 x_1} G_5$ (graphs₁, Box II) (*Step 2*).

Class 12 (Box V): internal preAUT maps



Graph data (lft structures in this box).

1. G_{19} :
 - underlying graph class = U_3 .
 - red vertex = x_3 .
 - red edge = $\{x_3, \bar{x}_2\}$.
 - purple edges ($|\mathcal{P}| = 9$):
 - $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 - $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 - $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
 - black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.
2. G_{24} :
 - underlying graph class = U_3 .
 - red vertex = $\bar{x}_4$.
 - red edge = $\{\bar{x}_2, \bar{x}_4\}$.
 - purple edges ($|\mathcal{P}| = 9$):
 - $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 - $P_2 : \{x_2, \bar{x}_3\}, \{x_2, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$
 - $P_3 : \{x_3, x_4\}, \{x_3, x_5\}, \{x_4, x_5\}$.
 - black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.
3. G_{25} :
 - underlying graph class = U_3 .
 - red vertex = $\bar{x}_5$.

- red edge = $\{\bar{x}_2, \bar{x}_5\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, \bar{x}_3\}, \{x_2, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_4\}$
 $P_3 : \{x_3, x_4\}, \{x_3, x_5\}, \{x_4, x_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

Arrows. Blue: Step 2 edge label. Green: permutation φ from ep-isomorphism (Step 3).

Maps represented (internal to Box V).

1. $G_{24} \xrightarrow{\varphi} G_{19}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto \bar{x}_4, x_4 \mapsto \bar{x}_3, x_5 \mapsto \bar{x}_5$ (permutation / ep-isomorphism).
2. $G_{25} \xrightarrow{\varphi} G_{19}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto \bar{x}_4, x_4 \mapsto \bar{x}_5, x_5 \mapsto \bar{x}_3$ (permutation / ep-isomorphism).

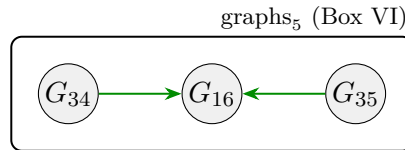
Incoming preAUT edges (into this box).

1. $G_{16} \xrightarrow{x_3 \rightarrow x_2 \ x_3} G_{19}$ (graphs₅, Box VI) (Step 2).
2. $G_{18} \xrightarrow{\bar{x}_4 \rightarrow x_2 \ \bar{x}_4} G_{24}$ (graphs₃, Box VII) (Step 2).
3. $G_{18} \xrightarrow{\bar{x}_5 \rightarrow x_2 \ \bar{x}_5} G_{25}$ (graphs₃, Box VII) (Step 2).

Outgoing preAUT edges (from this box).

1. $G_{19} \xrightarrow{x_1 \rightarrow x_3 \ x_1} G_{26}$ (graphs₁, Box II) (Step 2).
2. $G_{19} \xrightarrow{x_3 \rightarrow \bar{x}_1 \ x_3} G_{27}$ (graphs₉, Box IV) (Step 2).
3. $G_{19} \xrightarrow{x_3 \rightarrow x_1 \ x_3} G_{28}$ (graphs₉, Box IV) (Step 2).
4. $G_{19} \xrightarrow{\bar{x}_1 \rightarrow x_3 \ \bar{x}_1} G_{29}$ (graphs₁, Box II) (Step 2).

Class 5 (Box VI): internal preAUT maps



Graph data (lft structures in this box).

1. G_{16} :
 - underlying graph class = U_4 .
 - red vertex = x_3 .
 - red edge = $\{x_3, x_5\}$.
 - purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
 - black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

2. G_{34} :

- underlying graph class = U_4 .
- red vertex = x_1 .
- red edge = $\{x_1, x_4\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_2 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$
 $P_3 : \{\bar{x}_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_4\}, \{\bar{x}_2, \bar{x}_4\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

3. G_{35} :

- underlying graph class = U_4 .
- red vertex = $\bar{x}_1$.
- red edge = $\{x_4, \bar{x}_1\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_2\}, \{x_1, \bar{x}_4\}, \{\bar{x}_2, \bar{x}_4\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

Arrows. Blue: Step 2 edge label. Green: permutation φ from ep-isomorphism (Step 3).

Maps represented (internal to Box VI).

1. $G_{34} \xrightarrow{\varphi} G_{16}$, where $\varphi = x_1 \mapsto x_3, x_2 \mapsto x_4, x_3 \mapsto x_1, x_4 \mapsto x_5, x_5 \mapsto x_2$ (*permutation / ep-isomorphism*).
2. $G_{35} \xrightarrow{\varphi} G_{16}$, where $\varphi = x_1 \mapsto \bar{x}_3, x_2 \mapsto x_4, x_3 \mapsto x_1, x_4 \mapsto x_5, x_5 \mapsto x_2$ (*permutation / ep-isomorphism*).

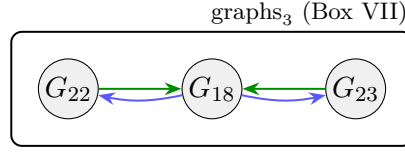
Incoming preAUT edges (into this box).

1. $G_{12} \xrightarrow{x_3 \rightarrow \bar{x}_5 x_3} G_{16}$ (graphs₇, Box X) (*Step 2*).
2. $G_{17} \xrightarrow{x_3 \rightarrow \bar{x}_5 x_3} G_{16}$ (graphs₁₀, Box IX) (*Step 2*).
3. $G_{31} \xrightarrow{x_1 \rightarrow \bar{x}_4 x_1} G_{34}$ (graphs₁₃, Box XI) (*Step 2*).
4. $G_{31} \xrightarrow{\bar{x}_1 \rightarrow \bar{x}_4 \bar{x}_1} G_{35}$ (graphs₁₃, Box XI) (*Step 2*).

Outgoing preAUT edges (from this box).

1. $G_{16} \xrightarrow{x_3 \rightarrow x_4 x_3} G_{12}$ (graphs₇, Box X) (*Step 2*).
2. $G_{16} \xrightarrow{x_4 \rightarrow x_3 x_4} G_{14}$ (graphs₇, Box X) (*Step 2*).
3. $G_{16} \xrightarrow{x_2 \rightarrow x_3 x_2} G_{18}$ (graphs₃, Box VII) (*Step 2*).
4. $G_{16} \xrightarrow{x_3 \rightarrow x_2 x_3} G_{19}$ (graphs₁₂, Box V) (*Step 2*).

Class 3 (Box VII): internal preAUT maps



Graph data (labeled structures in this box).

1. G_{18} :

- underlying graph class = U_4 .
- red vertex = x_2 .
- red edge = $\{x_2, \bar{x}_3\}$.
- purple edges ($|\mathcal{P}| = 9$):
 - $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 - $P_2 : \{x_3, x_4\}, \{x_3, x_5\}, \{x_4, x_5\}$
 - $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

2. G_{22} :

- underlying graph class = U_4 .
- red vertex = x_2 .
- red edge = $\{x_2, x_4\}$.
- purple edges ($|\mathcal{P}| = 9$):
 - $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 - $P_2 : \{x_3, x_4\}, \{x_3, x_5\}, \{x_4, x_5\}$
 - $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

3. G_{23} :

- underlying graph class = U_4 .
- red vertex = x_2 .
- red edge = $\{x_2, x_5\}$.
- purple edges ($|\mathcal{P}| = 9$):
 - $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 - $P_2 : \{x_3, x_4\}, \{x_3, x_5\}, \{x_4, x_5\}$
 - $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

Arrows. Blue: Step 2 edge label. Green: permutation φ from ep-isomorphism (Step 3).

Maps represented (internal to Box VII).

1. $G_{18} \xrightarrow{x_2 \rightarrow \bar{x}_4 \ x_2} G_{22}$ (Step 2).
2. $G_{18} \xrightarrow{x_2 \rightarrow \bar{x}_5 \ x_2} G_{23}$ (Step 2).
3. $G_{22} \xrightarrow{\varphi} G_{18}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto \bar{x}_4, x_4 \mapsto \bar{x}_3, x_5 \mapsto \bar{x}_5$ (permutation / ep-isomorphism).

4. $G_{23} \xrightarrow{\varphi} G_{18}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto \bar{x}_4, x_4 \mapsto \bar{x}_5, x_5 \mapsto \bar{x}_3$ (permutation / ep-isomorphism).

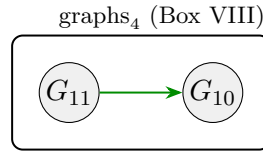
Incoming preAUT edges (into this box).

1. $G_{16} \xrightarrow{x_2 \rightarrow x_3 x_2} G_{18}$ (graphs₅, Box VI) (Step 2).

Outgoing preAUT edges (from this box).

1. $G_{18} \xrightarrow{\bar{x}_4 \rightarrow x_2 \bar{x}_4} G_{24}$ (graphs₁₂, Box V) (Step 2).
 2. $G_{18} \xrightarrow{\bar{x}_5 \rightarrow x_2 \bar{x}_5} G_{25}$ (graphs₁₂, Box V) (Step 2).

Class 4 (Box VIII): internal preAUT maps



Graph data (lft structures in this box).

1. G_{10} :
- underlying graph class = U_4 .
 - red vertex = x_3 .
 - red edge = $\{x_2, x_3\}$.
 - purple edges ($|\mathcal{P}| = 9$):
 - $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 - $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 - $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
 - black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.
2. G_{11} :
- underlying graph class = U_4 .
 - red vertex = $\bar{x}_2$.
 - red edge = $\{\bar{x}_2, \bar{x}_3\}$.
 - purple edges ($|\mathcal{P}| = 9$):
 - $P_1 : \{x_1, x_3\}, \{x_1, \bar{x}_1\}, \{x_3, \bar{x}_1\}$
 - $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 - $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
 - black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

Arrows. Blue: Step 2 edge label. Green: permutation φ from ep-isomorphism (Step 3).

Maps represented (internal to Box VIII).

1. $G_{11} \xrightarrow{\varphi} G_{10}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto \bar{x}_3, x_3 \mapsto \bar{x}_2, x_4 \mapsto \bar{x}_4, x_5 \mapsto \bar{x}_5$ (permutation / ep-isomorphism).

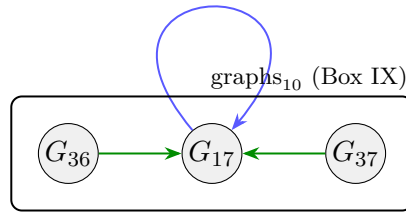
Incoming preAUT edges (into this box).

1. $G_6 \xrightarrow{x_3 \rightarrow \bar{x}_2 \ x_3} G_{10}$ (graphs₉, Box IV) (*Step 2*).
2. $G_6 \xrightarrow{\bar{x}_2 \rightarrow x_3 \ \bar{x}_2} G_{11}$ (graphs₉, Box IV) (*Step 2*).

Outgoing preAUT edges (from this box).

1. $G_{10} \xrightarrow{x_3 \rightarrow x_4 \ x_3} G_{12}$ (graphs₇, Box X) (*Step 2*).
2. $G_{10} \xrightarrow{x_3 \rightarrow x_5 \ x_3} G_{13}$ (graphs₇, Box X) (*Step 2*).
3. $G_{10} \xrightarrow{x_4 \rightarrow x_3 \ x_4} G_{14}$ (graphs₇, Box X) (*Step 2*).
4. $G_{10} \xrightarrow{x_5 \rightarrow x_3 \ x_5} G_{15}$ (graphs₇, Box X) (*Step 2*).

Class 10 (Box IX): internal preAUT maps



Graph data (lft structures in this box).

1. G_{17} :
 - underlying graph class = U_1 .
 - red vertex = $\bar{x}_5$.
 - red edge = $\{\bar{x}_3, \bar{x}_5\}$.
 - purple edges ($|\mathcal{P}| = 9$):
 - $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 - $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 - $P_3 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_4\}$.
 - black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.
2. G_{36} :
 - underlying graph class = U_1 .
 - red vertex = $\bar{x}_4$.
 - red edge = $\{x_1, \bar{x}_4\}$.
 - purple edges ($|\mathcal{P}| = 9$):
 - $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 - $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 - $P_3 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$.
 - black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.
3. G_{37} :
 - underlying graph class = U_1 .
 - red vertex = $\bar{x}_4$.
 - red edge = $\{\bar{x}_1, \bar{x}_4\}$.

- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

Arrows. Blue: Step 2 edge label. Green: permutation φ from ep-isomorphism (Step 3).

Maps represented (internal to Box IX).

1. $G_{17} \xrightarrow{\bar{x}_5 \rightarrow x_3 \bar{x}_5} G_{17}$ (Step 2).
2. $G_{36} \xrightarrow{\varphi} G_{17}$, where $\varphi = x_1 \mapsto \bar{x}_3, x_2 \mapsto x_4, x_3 \mapsto x_1, x_4 \mapsto x_5, x_5 \mapsto x_2$ (permutation / ep-isomorphism).
3. $G_{37} \xrightarrow{\varphi} G_{17}$, where $\varphi = x_1 \mapsto x_3, x_2 \mapsto x_4, x_3 \mapsto x_1, x_4 \mapsto x_5, x_5 \mapsto x_2$ (permutation / ep-isomorphism).

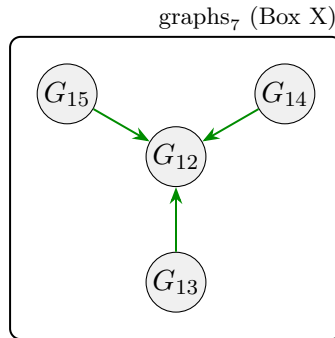
Incoming preAUT edges (into this box).

1. $G_{12} \xrightarrow{\bar{x}_5 \rightarrow x_3 \bar{x}_5} G_{17}$ (graphs₇, Box X) (Step 2).
2. $G_{31} \xrightarrow{\bar{x}_4 \rightarrow \bar{x}_1 \bar{x}_4} G_{36}$ (graphs₁₃, Box XI) (Step 2).
3. $G_{31} \xrightarrow{\bar{x}_4 \rightarrow x_1 \bar{x}_4} G_{37}$ (graphs₁₃, Box XI) (Step 2).

Outgoing preAUT edges (from this box).

1. $G_{17} \xrightarrow{x_3 \rightarrow \bar{x}_5 x_3} G_{16}$ (graphs₅, Box VI) (Step 2).
2. $G_{17} \xrightarrow{\bar{x}_4 \rightarrow \bar{x}_5 \bar{x}_4} G_{20}$ (graphs₈, Box XII) (Step 2).
3. $G_{17} \xrightarrow{\bar{x}_5 \rightarrow \bar{x}_4 \bar{x}_5} G_{21}$ (graphs₈, Box XII) (Step 2).

Class 7 (Box X): internal preAUT maps



Graph data (lft structures in this box).

1. G_{12} :
 - underlying graph class = U_1 .
 - red vertex = x_3 .
 - red edge = $\{x_3, \bar{x}_4\}$.

- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

2. G_{13} :

- underlying graph class = U_1 .
- red vertex = x_3 .
- red edge = $\{x_3, \bar{x}_5\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

3. G_{14} :

- underlying graph class = U_1 .
- red vertex = x_4 .
- red edge = $\{x_4, \bar{x}_3\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_3\}, \{x_2, x_5\}, \{x_3, x_5\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

4. G_{15} :

- underlying graph class = U_1 .
- red vertex = x_5 .
- red edge = $\{x_5, \bar{x}_3\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_3\}, \{x_2, x_4\}, \{x_3, x_4\}$
 $P_3 : \{\bar{x}_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_5\}, \{\bar{x}_4, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

Arrows. Blue: Step 2 edge label. Green: permutation φ from ep-isomorphism (Step 3).

Maps represented (internal to Box X).

1. $G_{13} \xrightarrow{\varphi} G_{12}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto x_3, x_4 \mapsto x_5, x_5 \mapsto x_4$ (permutation / ep-isomorphism).
2. $G_{14} \xrightarrow{\varphi} G_{12}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto x_4, x_4 \mapsto x_3, x_5 \mapsto x_5$ (permutation / ep-isomorphism).
3. $G_{15} \xrightarrow{\varphi} G_{12}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto x_4, x_4 \mapsto x_5, x_5 \mapsto x_3$ (permutation / ep-isomorphism).

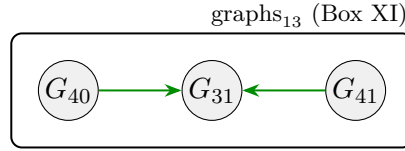
Incoming preAUT edges (into this box).

1. $G_{10} \xrightarrow{x_3 \rightarrow x_4 x_3} G_{12}$ (graphs₄, Box VIII) (*Step 2*).
2. $G_{10} \xrightarrow{x_3 \rightarrow x_5 x_3} G_{13}$ (graphs₄, Box VIII) (*Step 2*).
3. $G_{10} \xrightarrow{x_4 \rightarrow x_3 x_4} G_{14}$ (graphs₄, Box VIII) (*Step 2*).
4. $G_{10} \xrightarrow{x_5 \rightarrow x_3 x_5} G_{15}$ (graphs₄, Box VIII) (*Step 2*).
5. $G_{16} \xrightarrow{x_3 \rightarrow x_4 x_3} G_{12}$ (graphs₅, Box VI) (*Step 2*).
6. $G_{16} \xrightarrow{x_4 \rightarrow x_3 x_4} G_{14}$ (graphs₅, Box VI) (*Step 2*).

Outgoing preAUT edges (from this box).

1. $G_{12} \xrightarrow{x_3 \rightarrow \bar{x}_5 x_3} G_{16}$ (graphs₅, Box VI) (*Step 2*).
2. $G_{12} \xrightarrow{\bar{x}_5 \rightarrow x_3 \bar{x}_5} G_{17}$ (graphs₁₀, Box IX) (*Step 2*).

Class 13 (Box XI): internal preAUT maps



Graph data (lft structures in this box).

1. G_{31} :
 - underlying graph class = U_1 .
 - red vertex = $\bar{x}_4$.
 - red edge = $\{\bar{x}_2, \bar{x}_4\}$.
 - purple edges ($|\mathcal{P}| = 9$):
 - $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 - $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 - $P_3 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$.
 - black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.
2. G_{40} :
 - underlying graph class = U_1 .
 - red vertex = x_3 .
 - red edge = $\{x_3, \bar{x}_2\}$.
 - purple edges ($|\mathcal{P}| = 9$):
 - $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 - $P_2 : \{x_2, \bar{x}_3\}, \{x_2, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$
 - $P_3 : \{x_4, x_5\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_4\}$.
 - black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.
3. G_{41} :
 - underlying graph class = U_1 .
 - red vertex = $\bar{x}_3$.
 - red edge = $\{\bar{x}_2, \bar{x}_3\}$.

- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_3\}, \{x_2, \bar{x}_5\}, \{x_3, \bar{x}_5\}$
 $P_3 : \{x_4, x_5\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_4\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

Arrows. Blue: Step 2 edge label. Green: permutation φ from ep-isomorphism (Step 3).

Maps represented (internal to Box XI).

1. $G_{40} \xrightarrow{\varphi} G_{31}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto \bar{x}_4, x_4 \mapsto x_3, x_5 \mapsto \bar{x}_5$ (permutation / ep-isomorphism).
2. $G_{41} \xrightarrow{\varphi} G_{31}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto x_4, x_4 \mapsto x_3, x_5 \mapsto \bar{x}_5$ (permutation / ep-isomorphism).

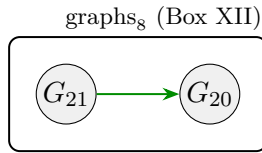
Incoming preAUT edges (into this box).

1. $G_{20} \xrightarrow{\bar{x}_4 \rightarrow x_2 \bar{x}_4} G_{31}$ (graphs₈, Box XII) (Step 2).
2. $G_{30} \xrightarrow{\bar{x}_4 \rightarrow x_2 \bar{x}_4} G_{31}$ (graphs₁₁, Box XIII) (Step 2).
3. $G_{32} \xrightarrow{x_3 \rightarrow x_2 x_3} G_{40}$ (graphs₁₄, Box XIV) (Step 2).
4. $G_{32} \xrightarrow{\bar{x}_3 \rightarrow x_2 \bar{x}_3} G_{41}$ (graphs₁₄, Box XIV) (Step 2).

Outgoing preAUT edges (from this box).

1. $G_{31} \xrightarrow{x_1 \rightarrow \bar{x}_4 x_1} G_{34}$ (graphs₅, Box VI) (Step 2).
2. $G_{31} \xrightarrow{\bar{x}_1 \rightarrow \bar{x}_4 \bar{x}_1} G_{35}$ (graphs₅, Box VI) (Step 2).
3. $G_{31} \xrightarrow{\bar{x}_4 \rightarrow \bar{x}_1 \bar{x}_4} G_{36}$ (graphs₁₀, Box IX) (Step 2).
4. $G_{31} \xrightarrow{\bar{x}_4 \rightarrow x_1 \bar{x}_4} G_{37}$ (graphs₁₀, Box IX) (Step 2).

Class 8 (Box XII): internal preAUT maps



Graph data (lft structures in this box).

1. G_{20} :
 - underlying graph class = U_2 .
 - red vertex = $\bar{x}_4$.
 - red edge = $\{x_5, \bar{x}_4\}$.
 - purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$.

- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

2. G_{21} :

- underlying graph class = U_2 .
- red vertex = $\bar{x}_5$.
- red edge = $\{x_4, \bar{x}_5\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_2, x_4\}, \{x_2, x_5\}, \{x_4, x_5\}$
 $P_3 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_4\}, \{\bar{x}_3, \bar{x}_4\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

Arrows. Blue: Step 2 edge label. Green: permutation φ from ep-isomorphism (Step 3).

Maps represented (internal to Box XII).

1. $G_{21} \xrightarrow{\varphi} G_{20}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto x_3, x_4 \mapsto x_5, x_5 \mapsto x_4$ (*permutation / ep-isomorphism*).

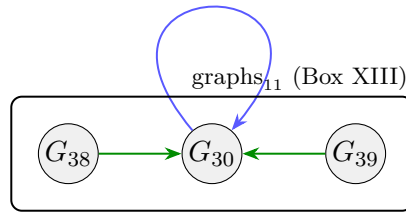
Incoming preAUT edges (into this box).

1. $G_{17} \xrightarrow{\bar{x}_4 \rightarrow \bar{x}_5 \bar{x}_4} G_{20}$ (graphs₁₀, Box IX) (*Step 2*).
2. $G_{17} \xrightarrow{\bar{x}_5 \rightarrow \bar{x}_4 \bar{x}_5} G_{21}$ (graphs₁₀, Box IX) (*Step 2*).

Outgoing preAUT edges (from this box).

1. $G_{20} \xrightarrow{x_2 \rightarrow \bar{x}_4 x_2} G_{30}$ (graphs₁₁, Box XIII) (*Step 2*).
2. $G_{20} \xrightarrow{\bar{x}_4 \rightarrow x_2 \bar{x}_4} G_{31}$ (graphs₁₃, Box XI) (*Step 2*).

Class 11 (Box XIII): internal preAUT maps



Graph data (lft structures in this box).

1. G_{30} :

- underlying graph class = U_2 .
- red vertex = x_2 .
- red edge = $\{x_2, x_4\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$
 $P_3 : \{x_4, x_5\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_4\}$.

- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

2. G_{38} :

- underlying graph class = U_2 .
- red vertex = x_2 .
- red edge = $\{x_2, x_3\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$
 $P_3 : \{x_4, x_5\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_4\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

3. G_{39} :

- underlying graph class = U_2 .
- red vertex = x_2 .
- red edge = $\{x_2, \bar{x}_3\}$.
- purple edges ($|\mathcal{P}| = 9$):
 $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 $P_2 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$
 $P_3 : \{x_4, x_5\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_4\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

Arrows. Blue: Step 2 edge label. Green: permutation φ from ep-isomorphism (Step 3).

Maps represented (internal to Box XIII).

1. $G_{30} \xrightarrow{x_2 \rightarrow \bar{x}_4 \ x_2} G_{30}$ (Step 2).
2. $G_{38} \xrightarrow{\varphi} G_{30}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto x_4, x_4 \mapsto x_3, x_5 \mapsto \bar{x}_5$ (permutation / ep-isomorphism).
3. $G_{39} \xrightarrow{\varphi} G_{30}$, where $\varphi = x_1 \mapsto x_1, x_2 \mapsto x_2, x_3 \mapsto \bar{x}_4, x_4 \mapsto x_3, x_5 \mapsto \bar{x}_5$ (permutation / ep-isomorphism).

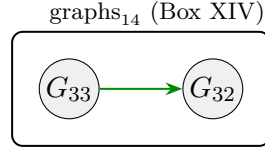
Incoming preAUT edges (into this box).

1. $G_{20} \xrightarrow{x_2 \rightarrow \bar{x}_4 \ x_2} G_{30}$ (graphs₈, Box XII) (Step 2).
2. $G_{32} \xrightarrow{x_2 \rightarrow \bar{x}_3 \ x_2} G_{38}$ (graphs₁₄, Box XIV) (Step 2).
3. $G_{32} \xrightarrow{x_2 \rightarrow x_3 \ x_2} G_{39}$ (graphs₁₄, Box XIV) (Step 2).

Outgoing preAUT edges (from this box).

1. $G_{30} \xrightarrow{\bar{x}_4 \rightarrow x_2 \ \bar{x}_4} G_{31}$ (graphs₁₃, Box XI) (Step 2).
2. $G_{30} \xrightarrow{x_2 \rightarrow x_5 \ x_2} G_{32}$ (graphs₁₄, Box XIV) (Step 2).
3. $G_{30} \xrightarrow{x_5 \rightarrow x_2 \ x_5} G_{33}$ (graphs₁₄, Box XIV) (Step 2).

Class 14 (Box XIV): internal preAUT maps



Graph data (lft structures in this box).

1. G_{32} :

- underlying graph class = U_2 .
- red vertex = x_2 .
- red edge = $\{x_2, \bar{x}_5\}$.
- purple edges ($|\mathcal{P}| = 9$):
 - $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 - $P_2 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$
 - $P_3 : \{x_4, x_5\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_4\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

2. G_{33} :

- underlying graph class = U_2 .
- red vertex = x_5 .
- red edge = $\{x_5, \bar{x}_2\}$.
- purple edges ($|\mathcal{P}| = 9$):
 - $P_1 : \{x_1, \bar{x}_1\}, \{x_1, \bar{x}_2\}, \{\bar{x}_1, \bar{x}_2\}$
 - $P_2 : \{x_2, x_4\}, \{x_2, \bar{x}_4\}, \{x_4, \bar{x}_4\}$
 - $P_3 : \{x_3, \bar{x}_3\}, \{x_3, \bar{x}_5\}, \{\bar{x}_3, \bar{x}_5\}$.
- black edges ($|\mathcal{B}| = 5$): $\{x_1, \bar{x}_1\}, \{x_2, \bar{x}_2\}, \{x_3, \bar{x}_3\}, \{x_4, \bar{x}_4\}, \{x_5, \bar{x}_5\}$.

Arrows. Blue: Step 2 edge label. Green: permutation φ from ep-isomorphism (Step 3).

Maps represented (internal to Box XIV).

1. $G_{33} \xrightarrow{\varphi} G_{32}$, where $\varphi = x_1 \mapsto x_3, x_2 \mapsto x_5, x_3 \mapsto x_1, x_4 \mapsto x_4, x_5 \mapsto x_2$ (*permutation / ep-isomorphism*).

Incoming preAUT edges (into this box).

1. $G_{30} \xrightarrow{x_2 \rightarrow x_5 \ x_2} G_{32}$ (graphs₁₁, Box XIII) (*Step 2*).
2. $G_{30} \xrightarrow{x_5 \rightarrow x_2 \ x_5} G_{33}$ (graphs₁₁, Box XIII) (*Step 2*).

Outgoing preAUT edges (from this box).

1. $G_{32} \xrightarrow{x_2 \rightarrow \bar{x}_3 \ x_2} G_{38}$ (graphs₁₁, Box XIII) (*Step 2*).
2. $G_{32} \xrightarrow{x_2 \rightarrow x_3 \ x_2} G_{39}$ (graphs₁₁, Box XIII) (*Step 2*).
3. $G_{32} \xrightarrow{x_3 \rightarrow x_2 \ x_3} G_{40}$ (graphs₁₃, Box XI) (*Step 2*).
4. $G_{32} \xrightarrow{\bar{x}_3 \rightarrow x_2 \ \bar{x}_3} G_{41}$ (graphs₁₃, Box XI) (*Step 2*).

E Working translation table: paper Roman boxes vs. code numbering

This table gives a practical lookup between the Roman-numeral boxes in the paper image and the internal numbering used by this project (Class # in the ltt-class appendix and graphs_k in preAUT/AUT generation).

Paper box	Class #	graphs_k
I	2	graphs_2
II	1	graphs_1
III	6	graphs_6
IV	9	graphs_9
V	12	graphs_{12}
VI	5	graphs_5
VII	3	graphs_3
VIII	4	graphs_4
IX	10	graphs_{10}
X	7	graphs_7
XI	13	graphs_{13}
XII	8	graphs_8
XIII	11	graphs_{11}
XIV	14	graphs_{14}